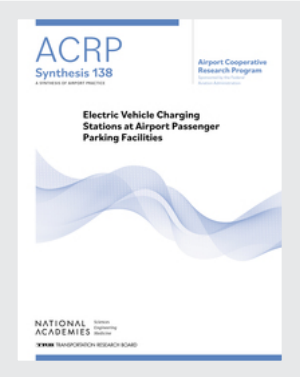
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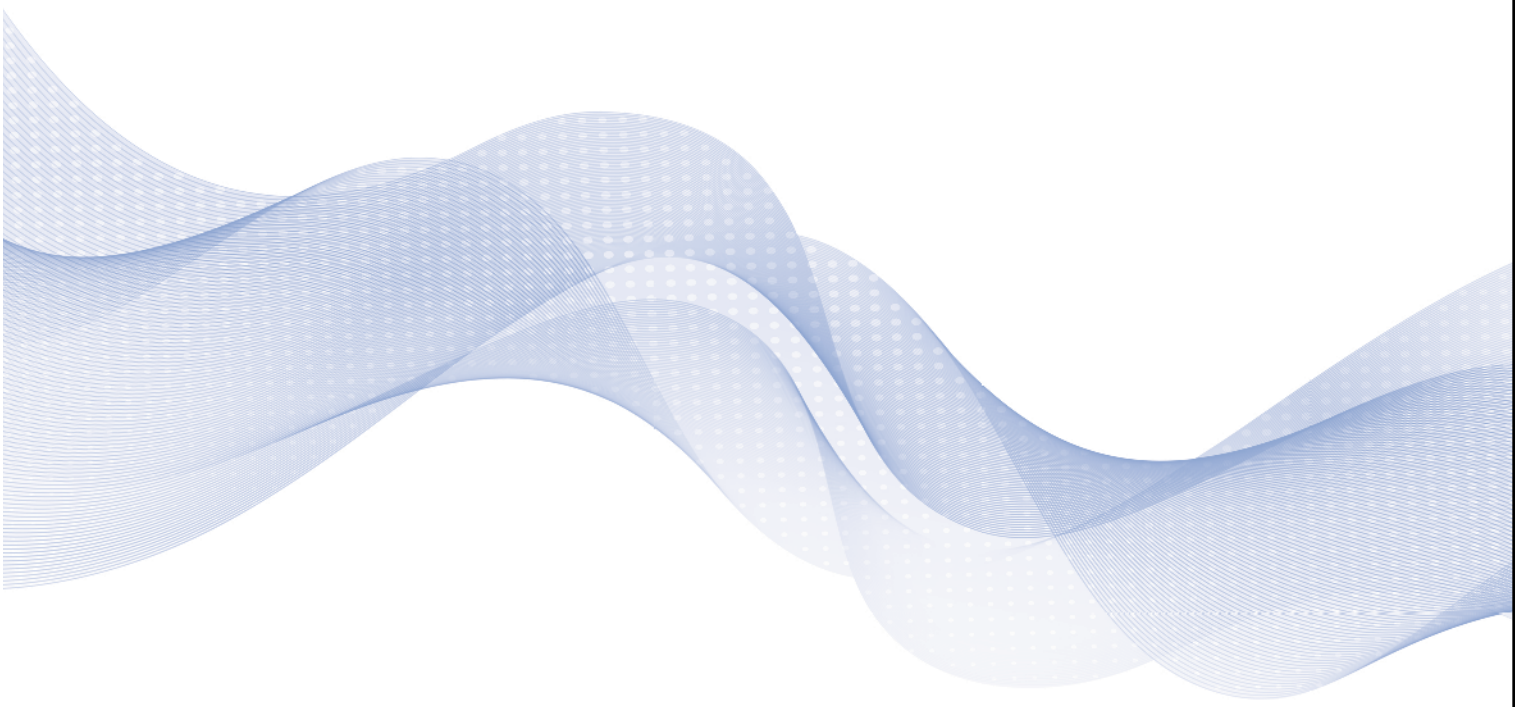
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ACRP
Synthesis 138
A SYNTHESIS OF AIRPORT PRACTICE

Airport Cooperative Research Program
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# Electric Vehicle Charging Stations at Airport Passenger Parking Facilities



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AIRPORT COOPERATIVE RESEARCH PROGRAM

ACRP SYNTHESIS 138

# Electric Vehicle Charging Stations at Airport Passenger Parking Facilities

*A Synthesis of Airport Practice*

**Jacqueline Kuzio**
**Harshit Shukla**
AND
**Karim Lujan**
**Texas A&M Transportation Institute**
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## AIRPORT COOPERATIVE RESEARCH PROGRAM

Airports are vital national resources. They serve a key role in transportation of people and goods and in regional, national, and international commerce. They are where the nation's aviation system connects with other modes of transportation and where federal responsibility for managing and regulating air traffic operations intersects with the role of state and local governments that own and operate most airports. Research is necessary to solve common operating problems, to adapt appropriate new technologies from other industries, and to introduce innovations into the airport industry. The Airport Cooperative Research Program (ACRP) serves as one of the principal means by which the airport industry can develop innovative near-term solutions to meet demands placed on it.

The need for ACRP was identified in *TRB Special Report 272: Airport Research Needs: Cooperative Solutions* in 2003, based on a study sponsored by the Federal Aviation Administration (FAA). ACRP carries out applied research on problems that are shared by airport operating agencies and not being adequately addressed by existing federal research programs. ACRP is modeled after the successful National Cooperative Highway Research Program (NCHRP) and Transit Cooperative Research Program (TCRP). ACRP undertakes research and other technical activities in various airport subject areas, including design, construction, legal, maintenance, operations, safety, policy, planning, human resources, and administration. ACRP provides a forum where airport operators can cooperatively address common operational problems.

ACRP was authorized in December 2003 as part of the Vision 100—Century of Aviation Reauthorization Act. The primary participants in the ACRP are (1) an independent governing board, the ACRP Oversight Committee (AOC), appointed by the Secretary of the U.S. Department of Transportation with representation from airport operating agencies, other stakeholders, and relevant industry organizations such as the Airports Council International-North America (ACI-NA), the American Association of Airport Executives (AAAE), the National Association of State Aviation Officials (NASAO), Airlines for America (A4A), and the Airport Consultants Council (ACC) as vital links to the airport community; (2) TRB as program manager and secretariat for the governing board; and (3) the FAA as program sponsor. In October 2005, the FAA executed a contract with the National Academy of Sciences formally initiating the program.

ACRP benefits from the cooperation and participation of airport professionals, air carriers, shippers, state and local government officials, equipment and service suppliers, other airport users, and research organizations. Each of these participants has different interests and responsibilities, and each is an integral part of this cooperative research effort.

Research problem statements for ACRP are solicited periodically but may be submitted to TRB by anyone at any time. It is the responsibility of the AOC to formulate the research program by identifying the highest priority projects and defining funding levels and expected products.

Once selected, each ACRP project is assigned to an expert panel appointed by TRB. Panels include experienced practitioners and research specialists; heavy emphasis is placed on including airport professionals, the intended users of the research products. The panels prepare project statements (requests for proposals), select contractors, and provide technical guidance and counsel throughout the life of the project. The process for developing research problem statements and selecting research agencies has been used by TRB in managing cooperative research programs since 1962. As in other TRB activities, ACRP project panels serve voluntarily without compensation.

Primary emphasis is placed on disseminating ACRP results to the intended users of the research: airport operating agencies, service providers, and academic institutions. ACRP produces a series of research reports for use by airport operators, local agencies, the FAA, and other interested parties; industry associations may arrange for workshops, training aids, field visits, webinars, and other activities to ensure that results are implemented by airport industry practitioners.

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Airport administrators, engineers, and researchers often face problems for which information already exists, either in documented form or as undocumented experience and practice. This information may be fragmented, scattered, and unevaluated. As a consequence, full knowledge of what has been learned about a problem may not be brought to bear on its solution. Costly research findings may go unused, valuable experience may be overlooked, and due consideration may not be given to recommended practices for solving or alleviating the problem.

There is information on nearly every subject of concern to the airport industry. Much of it derives from research or from the work of practitioners faced with problems in their day-to-day work. To provide a systematic means for assembling and evaluating such useful information and to make it available to the entire airport community, the Airport Cooperative Research Program authorized the Transportation Research Board to undertake a continuing project. This project, ACRP Project 11-03, "Synthesis of Information Related to Airport Practices," searches out and synthesizes useful knowledge from all available sources and prepares concise, documented reports on specific topics. Reports from this endeavor constitute an ACRP report series, Synthesis of Airport Practice.

This synthesis series reports on current knowledge and practice, in a compact format, without the detailed directions usually found in handbooks or design manuals. Each report in the series provides a compendium of the best knowledge available on those measures found to be the most successful in resolving specific problems.

## FOREWORD

By Jordan Christensen
Staff Officer
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Electric vehicles (EV) sales have increased significantly since the 2014 publication of *ACRP Synthesis 54: Electric Vehicle Charging Stations at Airport Parking Facilities*. Today, to meet passenger EV charging needs, more than 90 hub airports have installed EV charging in their passenger parking facilities. Given that passengers' needs for EV charging are expected to continue to rise, airports will benefit from understanding the challenges and opportunities related to providing EV charging in passenger parking facilities.

The objective of this synthesis is to describe how hub airports plan for, manage, and implement EV charging stations at public parking facilities. This synthesis includes research on (1) how airports determine the quantity, level, and location of charging stations, (2) how airports fund installation of charging stations, (3) ongoing operational and maintenance costs, and (4) safety concerns. The audience for this synthesis is airport planners, engineers, and managers.

This synthesis reflects information obtained through a literature review, a survey of airports, and interviews resulting in six case examples. Appendix C provides case examples from three small-hub, one medium-hub, and two large-hub airports that highlight the airports' experiences with EV charging infrastructure.

The Principal Investigator, Jacqueline Kuzio, and Harshit Shukla and Karim Lujan, all of Texas A&M Transportation Institute, College Station, TX, synthesized the information and wrote the report. The members of the topic panel are acknowledged on page iv. This synthesis is an immediately useful document that records the practices that were acceptable within the limitations of the knowledge available at the time of its preparation. As progress in research and practice continues, new knowledge will be added to that now at hand.

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SUMMARY

# Electric Vehicle Charging Stations at Airport Passenger Parking Facilities

Over the past decade, electric vehicle (EV) sales in the United States have increased from around 100,000 in 2013 to more than one million in 2023. In turn, airports have installed charging stations to meet their customers' needs, prepare for expected demand, and comply with local ordinances. In 2014, 37 airports were found to provide charging in some form at their airports. This synthesis effort found that at least 93 hub airports are offering EV charging in their passenger parking facilities. This rapid increase highlights a need to understand the state of the practice in terms of providing EV charging for passenger, or customer, use at airports. The changes in the electrified mobility space offer the opportunity to document challenges that remain as well as opportunities and innovative solutions to meet existing and future charging demand.

This synthesis documents the current experience with EV charging at airports, specifically within passenger parking facilities. Although the use of EVs and the need for charging equipment exists across the airport, publicly accessible charging is a unique use case that presents a different set of challenges than either employee parking or fleet charging facilities. Meeting the needs of airport passengers requires an understanding of typical parking times and, if EV charging is already installed, the ability to determine use patterns and potential needs. To assist airports, this synthesis documents the overall state of the practice and airport-reported lessons learned as well as providing information about charger types, locations, utility coordination and load management, funding and revenue issues, and safety.

This synthesis effort included a literature review, a survey, and interviews. Results from the survey and the interviews were developed into six case examples. The case examples included airports of various sizes from across the country and consist of three small-hub airports, one medium-hub airport, and two large-hub airports. The literature review involved a search of key transportation databases and a review of industry-leading reports, such as *ACRP Synthesis 54: Electric Vehicle Charging Stations at Airport Parking Facilities.* Using the results of the literature review, a set of survey questions was developed to address the following topics:

- EV Infrastructure at Airports
- Experience with EV Charging Installation
- Monitoring and Data Collection
- Funding for EV Charging Infrastructure
- Electrical Capacity, Utility Coordination, and Load Management
- Safety

The survey (provided as Appendix A) was sent to contacts at 123 hub airports in the United States. The survey had 37 responses, representing a response rate of 30%. From those responding to the survey, six airports were selected for more in-depth interviews. The



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2 Electric Vehicle Charging Stations at Airport Passenger Parking Facilities

interview framework used the same broad categories as the survey but sought more detail about the interviewees' experiences. The results of the interviews were used to develop case examples (provided herein as Appendix C) that provide information on current challenges and opportunities for airports installing EV charging equipment.

Understanding and meeting the needs of the airport passenger use case is a key topic and addresses questions about charging level, networking, and locations. Airports are in various stages of their EV planning, installation, and management and often do not know the effects of a specific charger or even a specific location until after installation. Pilot programs or trying out a charging solution can help airports gain the information and experience they need to develop these programs. Salt Lake City International Airport (SLC) used this approach and found it helpful in gaining the support of leadership and the facilities team who regularly work to install and maintain the equipment.

Although pilot programs are useful in the preliminary stages, it is important to consider the airport's full EV charging infrastructure needs so as to coordinate and forecast effectively in terms of the expected load requirement and budgetary needs. Portland International Jetport (PWM) did not use a coordinated strategy at first and now has different charging equipment, of varying quality, throughout their facilities. Moving forward, PWM plans to establish a policy on equipment quality. In addition, the airport wants to ensure that charging infrastructure will meet future demands, both in terms of passengers and technology.

Technological aspects of charging equipment and management can present a challenge to airports. Out of the 13 responding to that question, 10 indicated that their airport had networked chargers, thereby allowing their airports to collect data and monitor the assets. The interviewed airports had mixed opinions on the value of the data at their current level of EV charging infrastructure. Smaller airports thought that the costs outweighed the benefits but saw a need for such data if they were to expand their EV charging infrastructure. The cost of software solutions and managing the chargers were referenced as challenges at the large-hub airports. The interoperability of the software solution and the specific needs across the airport were important to airport staff.

Data and networking relate to utility coordination needs and concerns over electrical capacity. Most airports do not see providing Level 2 chargers as an issue with respect to load management or capacity. However, San Francisco (SFO) and SLC noted that, as their needs increase, load management could be a concern. In addition, all the airports noted that their challenges with electrical capacity revolved around the need for fast chargers. Given that fast chargers do not typically fit with the longer-term parking use cases at the airports, EV charging for passengers can be provided within existing capacity; however, small-hub airports had to evaluate their electrical panels to ensure capacity. Electrical capacity assessments provide valuable information both before and after installing the first set of chargers; however, one airport noted that an assessment would only be undertaken with a large expansion of EV chargers. Understanding when to conduct an assessment, coordinating with the utility, and managing load considerations requires communication among airport staff across departments.

Airport staff are also critical to providing charging stations for passenger vehicles. Airport staff who have not typically been part of the airport parking experience may be involved in these discussions. Airports of all sizes noted that their in-house electricians are important to the success and continued operations of equipment and that charging infrastructure maintenance and/or installation is an additional duty their facilities teams are now covering. In the interviews for the case examples, only one of the six airports specifically discussed workforce challenges. However, as the number of chargers at an airport increases, so

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does the importance of qualified and trained staff with the time to dedicate to maintaining equipment.

Staff training is important for safety because regular wear and tear can lead to safety and maintenance issues. Airports responding to the survey for this synthesis were following relevant codes and regulations—local, national, or both—and coordinating with their local fire departments to determine safety standards. Jacksonville International Airport (JAX) is planning to use a lithium-ion battery fire as their disaster exercise to assess their response capabilities. Overall, limited training has been provided or recommended, but airports do ensure their electricians are qualified and trained to work with the equipment. One case example noted that the local fire department has been proactive in pursuing training opportunities related to EVs. Outside of safety, basic staff training on handling of the equipment can reduce the need for replacement parts or other maintenance.

Although staff training is important, airports also spoke of the importance of a network and reaching out to those who have more experience or who can answer questions related to EV charging. Such networking and outreach can help provide an understanding of what worked and what did not for another airport. Airports are often dealing with a unique set of circumstances, but sharing knowledge can help other airports avoid similar challenges.

Further research that would support airports in addressing the remaining questions and challenges in terms of EV charging at their passenger parking facilities include

- Guidance on key decision points for EV charging infrastructure, including charger type and availability, when to select a networked charging asset, and location choices. Information on forecasting required electrical capacity and understanding load management, especially as such forecasting relates to decisions around charging equipment, would also help airports plan effectively.
- A guide, for interested parties, outlining the various management structures for airport EV charging. This guide could also review structures for private management and determine the departments that need to be involved in key decision areas.
- Workforce analysis to understand the effect of adding degrees of EV charging capacity as well as the training needs and opportunities.
- A review of safety requirements when installing and managing EV charging equipment in various scenarios (e.g., surface lots versus parking garages).

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CHAPTER 1

# Introduction

Transportation electrification is occurring across modes and sectors with implications for various parties, especially the airport community. Airports manage multiple modes of transportation, not just aviation. Airports connect to ground transportation through parking facilities, shuttles, and transit connections. Cargo operations may necessitate the accommodation of medium- and heavy-duty vehicles and promote close connections to both marine and rail terminals. In addition, airports use various medium- and light-duty vehicles to support their operations, both airside and landside. Although airports must understand all these electrification efforts, electric vehicle (EV) charging in passenger parking facilities is an area that airports have entered but about which they lack clear information. The objective of this synthesis is to explore the current state of the practice for airports regarding EV charging in their passenger parking facilities. EV charging for passengers or visitors represents a subset of the total electrification efforts but is a key area as EV sales in the United States (and therefore charging infrastructure needs) increase. Although EV charging installations for passenger use have implications beyond the parking facility (e.g., overall electrical capacity and load management as well as the increased workforce needs associated with new equipment), this synthesis focuses on the public-facing EV charging stations. This synthesis does not cover employee parking or other landside and airside charging needs for airports. Other charging use cases warrant their own exploration to determine the current state of the practice and airport experience.

Understanding the EV landscape today is critical to preparing airports for the subsequent electrification and charging needs in their parking facilities. In this report, EV is used as the broad category that encompasses both plug-in hybrid electric vehicles (PHEVs) and battery electric vehicles (BEVs). Because both types of EV can use charging equipment, this report reflects the airports' experience with all EVs. Although PHEVs may be less likely to use public charging facilities to top up their electric range, they can still access and use charging equipment.

*ACRP Synthesis 54: Electric Vehicle Charging Stations at Airport Parking Facilities* (Richard 2014) provided (1) a broad understanding of airports that offered charging, (2) a technology overview, and (3) a review of key business and policy decisions. Since the publication of *ACRP Synthesis 54*, EV sales have increased and states outside of the West Coast have adopted regulations to promote the transition to electric. Although information and guidance on EV charging has increased in recent years, there is little literature about charging needs at airports. Reports discuss airports in terms of fast-charging hubs, electric aviation, and energy management issues, but have not provided an overview of the airport experience with passenger EV charging. With increases in funding to support investment in charging equipment as well as charging providers willing to partner with commercial property owners, it is important to understand the broad array of challenges and opportunities related to EV charging and airport passenger parking facilities. This synthesis aims to address that gap with a state-of-the-practice review.



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In conducting the synthesis effort, the authors did a thorough a review of the literature, including relevant recent reports and guidance, which explored several key topics. These topics included

- **Charging infrastructure,** including the different charger types, the benefits of networked chargers, and how each charger fits into airport parking needs;
- **Charging management,** such as data collection and use, demand management, and effective siting;
- **Utility and energy issues,** which range from coordination requirements to load management and understanding electrical capacity;
- **Costs and revenue,** which are often closely associated (because cost can drive revenue decisions); costs and revenue address the range of investment costs as well as rate structures; and
- **Safety,** including the existing national standards and important codes and regulations.

The literature review informed the development of a survey sent to hub airports in the United States. The results of the survey provided an overview of the state of the practice for EV charging within passenger parking facilities. To gain a deeper understanding of the challenges and opportunities, the authors conducted interviews with six airports of varying size and geographic locations. In the interviews, airports presented their experiences with installing, managing, and maintaining charging equipment at their facilities.

This synthesis report is organized as follows:

- Chapter 1 provides an introduction and overview of the key topics covered in the synthesis.
- Chapter 2 presents an overview of the literature.
- Chapter 3 describes the hub airport experience in terms of the survey results, detailing key trends and findings for each topic.
- Chapter 4 covers more detailed information on the hub airport experience developed through the six case example interviews.
- Chapter 5 concludes the report with a summary of findings and areas for further research.
- References and a section on Terms and Abbreviations are provided.
- The full survey questionnaire and interview framework are provided in Appendix A. Appendices B and C provide detailed documentation of the survey results and the case example interviews, respectively.

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CHAPTER 2

# State of the Practice

*ACRP Synthesis 54: Electric Vehicle Charging Stations at Airport Parking Facilities* (Richard 2014) provided the first overview of EV charging at airport parking facilities, including the key challenges and opportunities that airports found through their experience with installation and management. In that synthesis, airports reported few to no issues with installation but were concerned with their needs in the future and how to accurately assess the number of charging ports required as well as appropriate locations. Airports also noted the wide range of costs associated with charging equipment, a concern that remains over a decade later. Key topics included technology, business and policy issues, and planning requirements. This chapter will provide an update on those topics, through a review of the literature, and discuss data collection and use policies, utility coordination and management, and safety. In addition, investment costs and funding opportunities will be explored, including an overview of innovative or emerging solutions being explored in the United States and internationally.

## The EV Landscape

The electrified mobility space is changing rapidly; in 2011, fully EV sales were 0.2% of the total car sales in the United States; in 2023, that rose to 7.6% and is expected to reach 10% by the end of 2024 (Muller 2024). Increasingly, consumers are selecting plug-in hybrids, with sales rising from less than 0.5% in 2014 to just over 2% in 2023 (Dwyer n.d.). The effect of these sales is that more and more of U.S. vehicle owners will have an EV and will expect to find charging when traveling, including at the airport. Alongside the increase in sales, federal funding to support charging infrastructure is in the middle of its rollout, which could lessen range anxiety (i.e., concern by a person driving an electric car that the battery will run out of power before the destination or a suitable charging point is reached). Allocated funding to support EVs and EV infrastructure at the federal level alone totals approximately $30.7 billion between fiscal year (FY) 2022 and FY2026 (Jellings 2021). Funding is available to support the use of clean vehicles, EV infrastructure, and grid-side investments.

The influx in funding, as well as the importance of the environment under the Infrastructure Investment and Jobs Act (IIJA), could help drive EV sales in the short term. In the long term, many manufacturers are increasing their EV production to meet corporate average fuel economy (CAFE) standards and emission reduction targets. Twelve states have adopted Advanced Clean Cars II standards (i.e., a regulation from California that stipulates all vehicle sales must be zero-emission for model year [MY] 2035 vehicles) (Khatib 2023). Some states have modified the regulation by adopting the sales percentages up to MY 2032. U.S. travelers have also indicated a strong interest in more sustainable travel, which could start with their ground transportation options (Virta 2023). Ultimately, access to charging will play a vital role in enabling the passenger fleet to switch to electric.



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The availability of, and access to, charging is moving from an amenity to a requirement in some areas as state and local governments adopt regulations or ordinances related to charging-capable, -ready, and -installed spaces. For example, the City of South Portland in Maine passed an ordinance in 2022 that requires 5% EV-installed and 20% EV-capable spaces in new or reconstructed parking facilities for non-residential, non-employee parking (City of South Portland 1992). Cities in California are adopting similar ordinances and the state has established a streamlined permitting process, which is legislatively mandated for many cities (California Governor's Office of Business and Economic Development n.d.). Requirements for charging in public parking facilities tend to focus on Level 2 charging, but California also has requirements for transportation network companies (TNCs) and airport shuttles to electrify, which may require fast chargers at the airport and in passenger parking facilities. Although these chargers may be installed by a third party or private entity, the airport will need to consider the additional load in their energy management for the airport.

Trends and regulations related to vehicle electrification require airports to quickly assess their current EV charging capacity and demand as well as their potential needs within parking facilities, especially long-term parking. Customers leaving their vehicles for multiple days or over a week may have a greater desire to charge their vehicles at a parking facility. Availability of charging is especially important to airport users who must travel greater distances to access the airport. In addition, short-term parking facilities could support fast-charging capabilities for passenger pickup and dropoff.

## Existing Electric Vehicle Infrastructure at Airports

*ACRP Synthesis 54: Electric Vehicle Charging Stations at Airport Parking Facilities* found that at least 37 airports in the United States were providing EV charging in some capacity (Richard 2014). Through the survey and a review of parking information available on airport websites, 93 hub airports now have at least one EV charger available for use by the public. The charging type or level typically depends on the type of parking facility as well as the size of the airport. Larger airports are more likely to have Level 3 or Direct Current Fast Charger (DCFC) chargers, given increased passenger throughput and the resulting demand for EV charging.

### Charging Equipment Levels

Charging equipment differs in terms of power and therefore time to charge. The differences are typically described in terms of "levels" and this section provides an overview of the three levels of charging and their key differences (see Table 1).

**Level 1.** The exact amount of charging capacity depends on components such as the size of the battery, the maximum power supplied to the grid, the charger, and the EV. Level 1 chargers take the longest to deliver a full charge but have universal capabilities. Level 1 chargers would be most beneficial for passengers who have long extended trips and use long-term parking that suits charging at a slower pace. A Level 1 charger uses a standard 120V outlet and a J1772 connector (which are used mainly in residential buildings) and provides 2 to 5 miles of range per hour of charging. This type of charging does not require any special equipment. To reach a full charge from empty takes approximately 5 to 6 hours for a PHEV and 40 to 50 hours for a BEV (USDOT n.d.a.).

**Level 2.** Level 2 chargers provide charging at a greater rate than Level 1—about 10 to 20 miles of range per hour of charge, depending on vehicle capability. Level 2 also requires a 208V to a 240V outlet. Level 2 chargers are commonly used in commercial settings but can support residential charging. Charging from empty to complete takes approximately 1 to 2 hours for a PHEV and 4 to 10 hours for a BEV (USDOT n.d.a.).

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**Table 1. Overview of charging equipment.**

| | Level 1 | Level 2 | DC Fast Charging |
|---|---|---|---|
| Connector Type | J1772 Connector  | J1772 Connector  | CCS Connector  CHAdeMO  NACS Connector  |
| Voltage | 120 V AC | 208 – 240 V AC | 400 V – 1000 V DC |
| Typical Power Output | 1kW | 7kW – 19kW | 50 – 350 kW |
| Estimated PHEV Charging Time from Empty | 5 – 6 hours | 1 – 2 hours | N/A |
| Estimated BEV Charge Time from Empty | 40 – 50 hours | 4 – 10 hours | 20 minutes – 1 hour |
| Estimated Electric Range per Hour of Charging | 2 – 5 miles | 10 – 20 miles | 180 – 240 miles |
| Typical Locations | Home | Home, Workplace, and Public | Public |

Source: U.S. Department of Transportation

**Level 3.** Level 3 charging or DCFC (i.e., the fastest form of EV charging) is most beneficial in short-term settings, such as along corridors. Level 3 charging is not universal to all electric vehicles, which adds a hurdle to its implementation. PHEVs currently do not work with fast-charging and can only use Levels 1 or 2. Level 3 charging requires 400V to 1000V DC plugs. Level 3 can provide such speed because it converts AC to DC in the charger instead of inside the car. This process allows Level 3 chargers to provide about 180 to 240 miles of range per hour, allowing an 80% charge from empty to take 20 minutes to an hour for a BEV. Three connectors can be used in DC fast-charging: Combined Charging System (CCS) connectors, CHAdeMO connectors, and the North American Charging Standard (NACS) connector are all available in this mode of charging (USDOT n.d.a.). CCS connectors use a combination of connectors to provide power up to 350kW; due to demand, there has also been implementation and testing of 400kW and 700kW CCS connectors. CHAdeMO, mainly used by Japanese-manufactured EVs, is a fast-charging connector. NACS connectors are also fast-charging-capable connectors recently adopted by most American vehicle manufacturers.

## Parking Use Cases for Different EV Chargers

Airport parking facilities present a challenge in terms of EV charging because the typical length of stay by airport users does not align with the current market for charging equipment. Level 2 chargers are widely available for commercial use because of their applicability for hospitality and retail use cases. However, airport customers tend to park in facilities for either a very short time (e.g., for pickup and dropoff) or for extended periods of time (e.g., days to weeks). For example, although only 30% of vehicles entering and exiting the airport stay for more than

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24 hours, vehicles parking for more than 24 hours occupy 70% of the parking spaces (Jacobs Consultancy et al. 2010). This example highlights the lack of daily or hourly parking that typically occurs in passenger parking facilities. The applications and use cases for Levels 1, 2, and 3 chargers at an airport are discussed in the following subsections.

**Level 1 and 2 Charging.** Although Level 1 and Level 2 chargers are common across most residential and commercial EV charging equipment today, Level 3 is typically reserved for high-traffic corridors where drivers require a fast turnaround. In terms of airport use cases, Levels 1 and 2 chargers would meet most airport passenger needs. Level 2 chargers are the most popular charging equipment in airports because Level 2 chargers better meet the needs of travelers (Carlton and Sultana 2022). Although a Level 2 charger can restore a vehicle to 80% charge within hours, users of airport parking facilities are often parked for days, rather than a few hours.

Although the needs of airport customers can be addressed with either Level 1 or Level 2 charging equipment, factors to consider when deciding on the appropriate mix of chargers include current demand, investment cost and budget, and the parking facility type and location (Virta 2023, Foster 2023). Airports with less throughput and more long-term parking may benefit from the reduced costs associated with Level 1 chargers. However, Level 1 chargers do not typically provide networking capabilities, thus limiting data collection opportunities. Existing charging infrastructure at airports is typically Level 2 equipment, but this may be driven by charging providers and other commercial use cases rather than airport needs (Richard 2014, Foster 2023).

**Level 3/DCFC Use Cases.** Level 3, or DCFC, chargers address a smaller proportion of passenger parking use cases at the airport but may serve short-term lots (e.g., cell phone waiting lots) or be needed to address ride-hail or taxi vehicles. Depending on the layout of the airport, ride-hail vehicles and taxis may use pickup and dropoff zones within existing passenger parking facilities. In such cases, airports would need to consider the charging needs of ride-hail vehicles and taxis along with charging equipment for airport customer use.

Although Level 3 chargers can be a significant investment, recent research has highlighted the needs and opportunities at airports that may be best served with "fast-charging depots." Fast-charging depots, also referred to as fast-charging hubs, are locations with Level 3 chargers that can support ride-hail vehicles and taxis, as well as airport pickup and dropoff. A hub or depot would have greater access to charging than a typical parking facility (e.g., more than 10 Level 3 chargers in one location). The continuous traffic turnover at large hubs, and even medium and small hubs, could support fast-charging stations near passenger pickup and dropoff areas (Foster 2023). Depending on location, these charging depots could also offer easily accessible public charging to non-airport customers. Many airports have developed cell phone lots to reduce dwell time near the terminal—these are places where Level 3 chargers could be installed and used to generate revenue. The Rocky Mountain Institute conducted a case study at Los Angeles International Airport (LAX) to understand the different business models as well as the time to recover the investment made by the airport (Carreon, Klock-McCook, and Mohanty 2022). The case study explored various business models to support charging, but each model saw LAX recover their investment within 10 years. The main use cases focused on ride-hail vehicles and taxis, but these pickup and dropoff zones can be within existing parking facilities at airports.

Although fast-charging depots could be used to generate revenue, challenges include

- The effect on overall electricity demand and capacity at the airports;
- The lack of data on the true cost of installation as well as operations and maintenance costs; and
- The need for utility upgrades and coordination (Carreon, Klock-McCook, and Mohanty 2022).

These challenges are not unique to Level 3 or fast-charging but are exacerbated by the increased electrical capacity needs as well as the utility demand charges that can occur from spikes in electricity

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usage at the airport. Airports have multiple facilities drawing from the grid, but this demand is often relatively flat across the day (Amarnath 2024). Introducing fast chargers would change the load profile and could increase costs if demand charges are incurred. The effect and importance of demand charges from a utility are discussed under Utility Coordination and Load Management.

## Challenges Associated with Charging at Airport Parking Facilities

Understanding the appropriate mix of EV charging based on parking facility need and airport use cases is a key challenge, and which is often related to charging asset location and understanding demand for charging. Using networked chargers allows the owner to collect data and answer questions related to use and demand, but using networked chargers requires an investment in the equipment and in staff time to monitor the equipment. Factors that affect charging purchase decisions, as well as the overall cost of ownership, include utility coordination and load management, capital equipment costs, installation and maintenance costs, and safety procedures.

### Charging Asset Location and Demand Management

Location of EV charging varies by airport with each airport having its own strategy. Airports generally balance customer preference with available infrastructure and safety considerations. From the customer perspective, it is important to keep in mind factors like visibility, accessibility, and use (Bonges and Lusk 2016). From the airport perspective, available electrical infrastructure (e.g., capacity at electrical panels) access and egress for safety purposes, and customer needs will play a role.

Given the unique use cases at airports, charging etiquette is also an important consideration. Locating charging equipment near facilities or in desirable spaces may increase the likelihood of a non-EV parking in the space. Selecting a location that provides good access to electrical infrastructure and is close but not directly in front of the terminal can ensure that the spaces are only used for charging. Signage and wayfinding for EV spaces also play roles in both promoting the amenity and proper charging etiquette (USDOE n.d.a., Caltrans n.d.). Public parking facilities must follow local codes and ordinances to ensure signs are enforceable. In addition, the *Manual on Uniform Traffic Control Devices* (MUTCD) guidelines should be followed, unless the parking facility is deemed private (USDOE n.d.a.).

Charger locations affect demand management as well. As EV sales increase, demand for charging at the airport may follow. Parking layouts that allow multiple vehicles to access the charging equipment can reduce issues with improper parking in an EV charging space and better use of the asset for longer stays in the facility. Policies and procedures around unplugging vehicles when their charge is complete would need to be considered if a charging owner is to maximize the use of the asset (Bonges and Lusk 2016). For example, this could be a service provided by staffed parking facilities. Many vehicles will not allow the charger to be decoupled when the vehicle is locked—this would limit the ability to unplug a vehicle outside of valet parking arrangements.

Ultimately, the airport may need to find a middle ground between customer preferences and ideal placement with respect to hard infrastructure. This tradeoff exists for most commercial parking facilities intending to install charging stations. Choosing a non-premium location can help ease demand—those who do not need to charge may choose to park elsewhere.

### Data Collection and Utilization Policies

A key component of demand management is understanding use patterns and electrical consumption at existing chargers or potential use based on survey responses or customer feedback.

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Collecting data to determine use practices typically requires a networked charging solution. Such data can then be incorporated into a dashboard for airport use; software solutions may be limited by or depend on charger brand as well. Advantages of networked chargers include

- The ability to assess a fee,
- Knowledge of the performance and maintenance needs of the asset,
- Understanding who is using the charger and for how long,
- Establishment of polices related to administration and access control, and
- The ability to determine energy limits or understand energy uses (Jones 2018, Young 2021, USDOT n.d.b.).

Selecting a charger that is part of a network can also be useful for marketing and informational purposes, given that the charger will appear on the charging network's map.

Non-networked chargers can reduce initial investment costs and lead to reduced maintenance costs. Depending on the solution chosen, the owner will typically need to purchase a software package to maintain access to the data. As airports with older charging assets consider upgrades to meet existing standards and airports plan for additional capacity; the decision will depend on whether the benefits of the data, in that format, outweigh the additional costs.

Ultimately, data collection can allow airports to establish performance metrics and targets, which may support accreditation programs or environmental assessments. In addition, a detailed understanding of usage and demand can be used to support grant applications or build partnerships with charging providers. In turn, understanding demand will help with electrical capacity planning and load management if more of the transportation sector electrifies, thus increasing the demand at airports.

## Utility Coordination and Load Management

A key theme when discussing EV charging installations and power requirements is to coordinate with the utility "early and often" (USDOT n.d.c., USDOT n.d.d.). Depending on the needs of the site, EV charging equipment might easily be installed within existing capacity; however, if large-scale installations or fast chargers are involved, this may not be the case. Communicating these needs with the utility as soon as they are known can reduce lead times on parts as well as heavy infrastructure upgrades, which tend to be costlier and could take years to complete. Therefore, ideally utility engagement and coordination begin as early as possible and the utility is kept informed of any plans or developments (USDOT n.d.d.). Forecasting load increases or growth in required capacity to support transportation electrification is essential to ensuring that the utility infrastructure is in place to address such growth.

Understanding future needs relies on an understanding of existing electrical capacity and use, even before EV charging equipment is installed. Electrical capacity assessments provide an overview of the maximum load a site can support across all facilities as well as an understanding of where electrical panel capacity exists. These assessments typically involve utility coordination to understand the potential for upgrades as well as existing maximum capacity to a site (Virta 2023, Richard 2014, van Bergen et al. 2021). Gathering data from charging assets provides the airport with an understanding of the impact to their overall electricity usage as well as providing information on potential future capacity needs for planning purposes.

Electrical capacity assessments can provide critical information on load management and where or when that may be required with relation to EV charging. The use of networked chargers allows an owner to establish a policy around energy usage to limit the effect of EV charging to overall power consumption and reduce load when necessary (Jones 2018). However, metering and submetering can be used to limit power usage by charging assets rather than investing in

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networked chargers. This is significant when considering the cost effects of demand charges from a utility company (Tucker and Snyder 2021). Some utilities offer rebates for shedding load during peak periods (i.e., a peak-time rebate) (Nandy, Botts, and Wenning 2022). The ability to limit power to non-essential functions, such as EV charging for airport customers, can support load-shedding. Most energy management models in the literature rely on revenue considerations or assume revenue generation for optimization (Firouzjah 2022; Babic et al. 2022; Faria, Baptista, and Farias 2014). Given that many airports do not assess a fee for EV charging, models may need to adopt other optimization protocols to better fit the airport's needs.

Because the adoption of charging infrastructure could cause a significant rise in the consumption of energy, research has explored the potential for a microgrid system to manage the various electrical demands of an airport, including EV charging (Guo et al. 2023). Airport parking lots have time-based rate structures and allow customers to book longer-term parking in advance. Long-term parking lots tend to have passengers that book their flights in advance with return tickets. This can allow the facility to have a database of demand predictions based on this information. Using vehicle-to-grid (V2G) technology, the parking lots could be used as aggregated energy storage units which would improve the stability of the grid system in the future (Guo et al. 2023; Tan, Kang, and Zhong 2023). V2G is not fully developed for large-scale use and may not be practical depending on utility policies or regulations. Copenhagen Airport is working with several partners to test the potential for V2G and smart charging to reduce emissions and electricity usage at the airport. By installing additional charging infrastructure, the airport can tap into the vehicles as an additional power source when needed (Thingvad 2023). Although this technology could help with load management in the future, it would incur additional investment costs.

## Investment Costs

The cost of installing, managing, and maintaining charging equipment varies greatly and depends on the charger level, whether the equipment is networked, and the need for electrical or utility upgrades. Cost elements can include

- Charging equipment,
- Transformers,
- Other utility upgrades (if necessary),
- Data contracts,
- Network contracts, and
- Payment systems (Nelder and Rogers 2020).

Those cost elements do not include the "soft costs" associated with installing charging equipment including costs for process/ing, marketing, and permitting.

Although the literature states that the cost of hardware elements is declining and assumed to continue declining, the costs associated with installation remain highly variable (Smith and Castellano 2015; Gamage, Tal, and Jenn 2023). For example, installing a Level 2 charger can cost anywhere from less than $1,000 to more than $10,000. The actual equipment costs can range from $1,000 to $5,000. In terms of Level 3, installation costs can range from $5,000 to $50,000; equipment costs can range from $10,000 to $40,000 (Nelder and Rogers 2020, Smith and Castellano 2015). In addition, the cost of managing the charging (e.g., monitoring usage and data collection and assisting customers) and the cost of maintaining the asset is not well documented. Estimates ranged from $250 a month to $400 a year (Frode, Lee, and Sahdev 2023; USDOE n.d.b.). However, as noted for hardware, costs are highly variable and can depend on usage of the equipment. It is also important to consider the fee that would need to be assessed to recover these costs if the intention is to cover costs through fees to the user. As with installation costs, maintenance costs depend on the type of charger and whether it is networked.

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When planning for EV charging equipment, it is important to consider the total cost of both installation and operations and maintenance. Although economies of scale can be realized, initial charging installations may require significant staff involvement. These internal soft costs can continuously decrease as future chargers are installed (Richard 2014). Involving all relevant personnel (from parking managers to electrical engineers) in regular discussions can ease this process and help to reduce time and cost.

Ultimately, the cost of charging equipment and lack of standardization in the charging industry can restrict growth (Foster 2023). The current availability of incentives and grant funding may play a critical role in enabling EV charging to be installed in parking facilities (van Bergen et al. 2021; Frode, Lee, and Sahdev 2023). As the industry grows, standards may be developed that reduce cost fluctuations and allow for better planning and budgeting regarding EV charging infrastructure.

## Safety

Standards and regulations relating to the safety of EV charging equipment are in varying stages of development, but key principles still apply. Guidance typically emphasizes the quality and rating of equipment, monitoring and maintenance, and specific guidelines when EVs may be parking or charging near internal combustion engine (ICE) vehicles or within a parking garage or covered facility (Durante 2023, Binder 2023). This section will provide an overview of guidance related to EV charging equipment as well as some key standards and codes that parking facility owners and managers should reference.

Basic safety provisions include using equipment rated for the intended use (e.g., under cover/indoor or outdoor) and ensuring that the manufacturer and parts are certified before installation (FEMA and U.S. Fire Administration n.d., NFPA 2023, Majano n.d.). Also, the asset needs to be monitored for excessive wear and tear that could cause safety issues. Finally, EV battery fires are different from ICE vehicle fires. When a lithium-ion battery fails, it goes through a process called "thermal runaway," where a single battery cell failure can cause the production of not just heat and oxygen but flammable and toxic gases. In turn, the fire can spread rapidly, burn longer, and even lead to explosions (O'Connor 2022). Although these events are relatively rare, unique safety considerations are required when EVs are parking within garages or near other vehicles.

The 2022 NFPA 13, *Standard for the Installation of Sprinkler Systems,* increased the recommended hazard classification for parking structures (O'Connor 2022). In 2023, NFPA 88A required all parking garages to have sprinkler systems installed in accordance with NFPA 13 (O'Connor 2022). As of early 2024, the National Fire Protection Association (NFPA) had few requirements for parking garages equipped with EV charging. The National Electrical Code (NEC), the standard for all electrical hazards, should be referenced before any installations take place.

Local codes and ordinances may have different standards and should always be referenced in conjunction with the requirements from NFPA and NEC. For example, the San Francisco Fire Department (SFFD) updated their guidance on sprinkler protection requirements of parking spaces associated with EV charging stations. The update was in response to 2022 NFPA 13, which SFFD did not believe adequately addressed the risks associated with EV fires (SFFD n.d.). SFFD is requiring (1) a fire sprinkler system designed per Extra Hazard Group II (EH2) for new buildings and (2) that existing buildings augment their sprinkler systems to meet EH2 standards. The requirement does not include an extension of 15 feet beyond the permitted parking spaces, as currently required by 2022 NFPA 13 Section 19.1.2(1) for the EH2 classification (SFFD n.d.).

EV charging equipment owners can also access free or paid training for their employees through the NFPA. The NFPA has 12 EV training resources covering guidance for emergency

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responders, installers, code officials, utilities, and vehicle technicians, and specific training on NFPA and NEC requirements (NFPA n.d.). All these training resources are offered online.

Overall, safety guidance is evolving for EV charging equipment as existing assets are analyzed and assessed. Regular maintenance and inspection are critical to safety because damaged or defective equipment can pose a hazard to technicians and the public. Employees who will be interacting regularly with the equipment should receive the appropriate training and education. Electricians must stay current on the codes, regulations, and guidance. Table 2 provides an overview of key codes and standards but is by no means an exhaustive list. Finally, integrating systems for fire protection into parking facilities can assist with responding to an EV fire (Binder 2023). These systems can include remote power boxes to cut off power at the charger when a fire is detected, fire detection systems close to the EV charging stations, and security cameras to confirm a fire incident.

**Table 2. Key codes and standards table.**

| Type | Title | Description |
|---|---|---|
| Standard | NFPA 13, Standard for the Installation of Sprinkler Systems – 2022 Edition | Suggests Ordinary Hazard 2 category for parking garages. Increases the required water density. |
| Code | NFPA 70, National Electrical Code (NEC) – 2021 Edition | Article 625 has a comprehensive list of requirements for EV charging stations.<br>Article 220 requires that load calculations consider EV supply equipment. |
| Standard | NFPA 88A, Standard for Parking Structures – 2023 Edition | Requires EV charging stations and equipment to be UL listed.<br>Requires an automatic sprinkler system in all parking garages, whether open or enclosed. |
| Code | International Building Code (IBC) – 2021 Edition | Sprinkler systems required for open parking garages exceeding 48,000 square feet. |
| Guidance | FM Global Property Loss Prevention Data Sheet (FMDS) 3-26 | Broad guidance for parking garages but elevated the hazard category from Hazard Category 2 (HC2) to Hazard Category 3 (HC3). Recognizes the increased risk for parking garages accommodating EVs. |
| Standard | Underwriters Laboratories (UL) 2202 | Standard addresses the specific safety requirements for EV charging equipment. UL conducts comprehensive testing of equipment to ensure it meets safety standards. |

Sources: Durante 2023, Binder 2023

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### Cybersecurity

Cybersecurity is a critical issue for EV charging, especially as networks expand across the United States (USDOE 2024a). The federal government is supporting research to test vulnerabilities and provide guidance on a risk-based approach to managing cybersecurity challenges that occur with EV charging infrastructure (USDOE 2024b; Grayson et al. 2023). However, limited research, standards, or best practices for EV charging equipment exist.

The research currently being funded by the federal government includes

- An EV Secure Architecture Laboratory Demonstration
- Improving secure communications through proper implementation of public key infrastructure (PKI)
- An assessment and coordination of EV supply equipment cybersecurity standards (USDOE 2024b).

Despite the lack of existing standards, guidance is available on conducting a threat assessment as well as risks to consider with EV charging infrastructure (Johnson et al. 2022). In addition, the research underway should support the development of standards and improve the consistency of their implementation. Ensuring safe and secure charging requires a full understanding of the risks, so selection, installation, and maintenance of chargers with internet connectivity should be coordinated with the IT department to address cybersecurity-related concerns.

## Opportunities to Support Charging at Airport Parking Facilities

Although airports need to consider several factors and challenges related to EV charging, opportunities exist to support charging at airport parking facilities. Recent federal legislation has expanded the availability of funding to support EV charging, and existing funding programs under the FAA can be used to support EVs and charging stations. This section will (1) provide an overview of the available federal programs and highlight other incentives, such as utility incentive programs; (2) discuss considerations for revenue recovery or assessing a fee; and (3) highlight some innovative solutions for EV charging at airports.

### Available Funding and Incentives

The FAA has developed several programs to support environmental initiatives using Airport Improvement Program (AIP) funds. The federal Bipartisan Infrastructure Law (BIL), which included several funding programs to support EV charging, can be used by airports. Grant funding may also be available at the state and local level; for certain grant funding programs, it is important to coordinate with state and local partners to use these funds effectively. Table 3 provides an overview of some of the funding programs available to airports.

Discretionary grant programs will require grant applications, and each program has its own requirements and selection criteria, so time and resources will be needed from the airport. Match requirements differ but are generally 20% for most federal funding programs. These grant programs are very competitive and generally receive many more applications than there is funding available. Programs such as VALE allow the airport to use funding to finance asset purchases.

Small-hub airports typically receive a greater share of AIP funds because small-hub airports represent 55% of all hub airports in the United States (NYSERDA n.d.). Allocation of funds through AIP depends on factors such as how much funding is authorized for the program, the number of enplanements, and the passenger facility charge assessed by the airport (FAA 2023b).

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16 Electric Vehicle Charging Stations at Airport Passenger Parking Facilities

**Table 3. Federal funding for EV charging equipment.**

| Program | Description | Notes | Agency |
|---|---|---|---|
| Airport Zero-Emissions Vehicle (ZEV) and Infrastructure Pilot Program (FAA n.d.a.) | Allows airports to use AIP funds to purchase ZEVs and to construct or modify the infrastructure needed to use ZEVs. | Any public use airport is eligible but favorable consideration is given to airports within EPA-designated non-attainment or maintenance areas. | FAA |
| Voluntary Airport Low Emissions Program (VALE) (FAA n.d.b.) | VALE allows airports to use AIP and passenger facility charges (PFCs) to finance the purchase of low-emission vehicles, refueling and recharging stations, gate electrification, and other airport air quality improvements. | Recently expanded to include all commercial service airports by FAA Reauthorization Act of 2024 [P.L. 118-63]. | FAA |
| Airport Improvement Program (AIP) Supplemental Grant Program (FAA n.d.c.) | Supplemental funds available for discretionary grants authorized under the AIP statute, 49 USC Chapters 471 and 475. | $3.18 billion has been made available for FY2023-2025. FAA has announced grant funding for a range of projects including electrification (FAA 2023). | FAA |
| Charging and Fueling Infrastructure Discretionary Grant Program (CFI) (USDOT-FHWA n.d.) | Two programs; Community Charging and Fueling Grants (Community Program); and (2) Alternative Fuel Corridor Grants (Corridor Program). | Eligible entities include units of local government, special purpose districts or public authorities with a transportation function, and state or local authorities with ownership of publicly accessible transportation facilities. | USDOT - FHWA |
| National Electric Vehicle Infrastructure (NEVI) (Rogers and Shepherd 2022) | Formula and discretionary funds available to support the build out of a national charging network along alternative fuel corridors (AFCs). | Once the network is built out along AFCs, funding can be used for other strategic charging locations. | USDOT-FHWA |
| Alternative Fuel Vehicle Refueling Property Tax Credits | Refueling property that is used to store or dispense clean-burning fuel or recharge electric motor vehicles. | As of January 1, 2023, the credit for qualified refueling property subject to depreciation equals 6% with a maximum credit of $100,000 for each single item of property. | IRS |

Note: These programs were available when this research was conducted (May 2024) but may no longer have funds available.

A U.S. Government Accountability Office (GAO) report in 2020 indicated that AIP funds did not cover the total needs at airports. In terms of EV charging equipment or planning, most airports have dedicated their AIP funds to other needs and explore alternative options to support these projects (GAO 2020). Seattle-Tacoma International Airport used AIP funding to conduct research on new approaches and test design strategies related to EV charging (Carreon, Klock-McCook, and Mohanty 2022; LeighFisher and Synergy Consultants 2018). Using AIP funds to conduct research can help airports make key decisions that will enable airports to install, manage, and maintain charging assets within their budgetary constraints.

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State and local governments also have grant funding opportunities to support EV infrastructure investments; however, these opportunities are also competitive and have limited funds. Although it is not possible to detail all the grant opportunities at the state and local level in this synthesis, examples include

- California's Charging Infrastructure for Government Fleets through the Clean Transportation Program. Up to $30 million is available to support EV charging infrastructure for light-duty government fleets (California Energy Commission n.d.).
- Colorado's "Charge Ahead Colorado" provides funding for Level 2 and 3 charging stations. The program provides a maximum funding of 80% of the total cost up to certain thresholds that depend on the type of charging (Colorado Energy Office n.d.).
- Denver International Airport used this program to help install ten new EV charging stations (DIA 2015).
- New York's Charge Ready NY offers rebates to install Level 2 EV charging stations at workplaces, multi-unit dwellings (MUDs) or public facilities owned and operated by municipal or state government entities. Up to $4,000 per charging station is available for public installations and up to $2,000 per charging station for workplace and MUDs (NYSERDA n.d.).
- The City of South Portland, Maine, has a Level 2 Electric Vehicle Charging Grant to support the installation of Level 2 chargers (City of South Portland n.d.). Priority is given to specific locations, including workplaces and businesses with heavy traffic.

Outside of grant funding, private investment and utility funding opportunities can be used to install and maintain charging equipment. Charging providers are installing stations across the country to meet demand, and if demand exists at an airport, there may be an opportunity to direct the investment (Joint Office of Energy and Transportation 2023). In addition, utility companies are offering various incentives, from grants to rebates, to support charging infrastructure (Allen et al. 2017, EV Charging Summit 2022). Utility incentives can scale with the type of charger, reducing the high initial costs of these investments.

Although not directly related to funding, certain accreditation programs award points based on the provision of EV charging equipment or for make-ready upgrades. For example, Leadership in Energy and Environmental Design (LEED) certification guidelines offers one point when a new facility installs EV charging in 5% of the facility or at least two spaces (whichever is greater). In addition, one point is offered when 10% or at least six spaces, (whichever is greater) are made EV-ready (U.S. Green Building Council n.d.). The Airport Carbon Accreditation program includes alternative fuel vehicles and charging stations within the strategies that can be used to reach Levels 2 and 3 of the program (Airport Carbon Accreditation n.d.). Airports with sustainability goals may be able to justify investments based on progress toward accreditation standards.

## Charging for "Charging" and Other Revenue Considerations

Before determining an effective rate for charging, EV charging owners need to determine whether assessing a fee is worthwhile. Although generating revenue from the charging asset can help to recover costs, the amount of charging installed, the costs associated with establishing a rate and collecting payment, and the potential total cost of ownership are important considerations.

In general, there are two primary business models for EV charging site hosts: owner-operator and third-party-owned and -operated. With the owner-operator business model, the site host is responsible for the installation, operation, and maintenance and can assess a fee. (In a third-party business model, the site host typically does not play a role in installation, operation, and maintenance but may assess a fee for use of land or space within a facility.) Typical pricing structures include no fee, a nominal fee to cover costs, and as a profit center (Atlas Public Policy 2020). Revenue generated from the use of the charging equipment can be used to help to recover costs associated with either the initial investment, maintenance costs, or both.

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The ability to assess a fee is typically easier with networked chargers, but networked chargers increase the cost to the owner in terms of the equipment and managing a payment system. Another option would be a valet or a gated area for EVs to determine usage and charge either a set or time-based fee to those using the chargers. A "charging as a service" model reduces the burden on the facility owner, but also reduces the potential for revenue recovery (Frode, Lee, and Sahdev 2023). Ultimately, if maintenance costs are low, the cost of determining a rate and collecting payment may outweigh the revenue generated.

Factors to consider when determining an appropriate rate include

- Data requirements to determine the rate based on price structure,
- Payment collection and management,
- Type of fee (e.g., fixed, time-based, or per-kWh), and
- How use of the asset will be monitored.

Airport parking facility goals are typically to generate revenue, enhance the customer experience, and improve efficiency (Mandle and Alandou 2022). Providing EV charging helps to improve the customer experience, but as this demand grows, the importance of generating revenue to keep up with costs may increase.

Understanding the total cost of ownership is beneficial when establishing a pricing structure. However, depending on the age of existing assets, not all EV charging equipment owners have the data required to conduct a total cost analysis. As experience with charging equipment grows, more data and information may be available to support pricing and revenue recovery decisions.

## Innovative Solutions

Various airports have created innovative solutions to address the challenges of increasing charging availability. These innovative solutions include smart parking upgrades (at LAX), mobile charging through ZiGGY (at DFW), and the development of a charging hub (at London's Gatwick Airport). These examples are discussed below.

**Smart Parking.** Smart parking refers to technology that improves efficiency or access to parking facilities. LAX Economy parking was awarded the "innovative facility of the year" award in 2022 because of LAX's success in modernizing and digitizing the parking experience. Updates, such as a mobile booking system, have provided customers with an efficient experience while providing the airport with new revenue opportunities (PowerFlex 2021). LAX's automated parking infrastructure includes (1) automatic gates with automated license plate readers (Pannunzio 2020) and (2) robots that go around the parking lot scanning license plates and ensuring that people are parked in the correct spots for their vehicle and parking type (ABM n.d.). LAX's smart parking initiative includes increasing access to EV charging through 500 Level 2 EV chargers across their parking as of 2021. Guests reserve their EV spots and then pay their EV fees through a one-step payment for parking and a fee based on kilowattage used (PowerFlex 2021).

**Mobile Charging.** DFW has partnered with EV Safe Charge to evaluate EV Safe Charge's mobile electric charging station, ZiGGY, in a public space. ZiGGY is a charging robot that "meets" drivers at their spaces rather than requiring EV owners to park next to physical charging infrastructure. DFW's VP of Innovation states that the airport is aware of the role EVs will play in the future. DFW is embracing this technology in hopes of meeting the demands of the future. The goal of ZiGGY is to speed up the entrances and exits of travelers using charging stations and allow charging to be offered anywhere in a facility. ZiGGY is deployed directly to the parking lot in which the vehicle is located and is then plugged in by the user. This technology provides Level 2 charging speeds, but there are plans to increase it to Level 3 soon. ZiGGY also benefits

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the facility in that it provides space on the charger for electronic advertising, which could provide an additional revenue stream (Paul 2023).

**Charging Hubs.** London Gatwick Airport is the first international airport to open an Electric Forecourt—essentially, an EV charging hub similar to a gas station. The Forecourt has 30 high-power and low-power EV chargers that are available for a fee. The fastest on-site charger can add up to 100 miles of range in 10 minutes. Other amenities are offered (e.g., a lounge area, Wi-Fi, a coffee shop, and a convenience store). The charging hub is in a strategic location frequented by millions of passengers, commuters, staff, businesses, and residents every year (*International Airport Review* 2024). Charging hubs can help address equity issues related to charging. Access to home charging can be limited, especially in dense urban environments (Carlton and Sultana 2022). London Gatwick's ambition is to achieve a rate of 60% zero or ultra-low-emission trips to and from the airport by 2030 (*International Airport Review* 2024). To ensure the growth of the EV user base, it is crucial to increase non-residential destination charging. Services such as the ZiGGY mobile charging station or the Electric Forecourt at London Gatwick provide users with a more accessible and convenient experience (Carlton and Sultana 2022).

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CHAPTER 3

# Overview of Survey Results

This chapter presents an overview of the results of the survey sent to hub airports in the United States. An online survey (see Appendix A, Survey Questionnaire options and Interview Framework) was developed based on the literature review of EV charging facilities at airport parking facilities. The survey was then distributed to 123 hub airports. In total, 36 airports responded to the survey, either through the online platform or via email. Highlights of the responses are presented in this chapter and detailed documentation on the survey and results is provided in Appendix B, Documentation of Survey Results and Analysis.

## Charging Stations and Associated Infrastructure

A review of the websites of all the hub airports in the United States along with the survey results found 93 of the hub airports (approximately 64% of all the hub airports) reported EV charging available in at least one of their passenger parking facilities. In terms of hub size, 46% of small-hub airports, 74% of medium-hub airports, and 100% of large-hub airports reported or advertised charging stations in their passenger parking facilities. (Note: airports may provide access to EV charging but not advertise this on their websites.)

Twenty-two survey respondents indicated that they had installed EV charging equipment. Airports are installing EV charging as an amenity, to meet customer expectations or expected demand, and to comply with local ordinances. Out of the 14 respondents without EV charging equipment, 8 indicated that they have plans to install charging equipment in the future. Of the airports without plans to install charging equipment, some indicated that they were exploring options outside of their parking facilities while others were intending to make their new parking facilities EV-ready but not EV-installed.

Key barriers to installing charging equipment were cost and infrastructure requirements. Other barriers included demand forecasting and management challenges, funding for maintenance, and the lack of onsite utility capacity to support charging.

### Different Charging Needs at Parking Facilities

Installing charging equipment at airports has occurred for various reasons; however, most airports indicated that offering charging as an amenity was important. Other factors included anticipated demand and responding to customer requests for charging.

The location of charging stations varied across airports with some airports focusing on charging availability in one key facility (e.g., their central garage) and other airports installing a couple of chargers in each of their parking facilities. To maximize the use of the asset, a few airports have chosen to provide charging in their valet parking. This arrangement allows the airport to charge more



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vehicles with less equipment and reduce wear and tear by individual customers. Within the facility, charging equipment location typically depends on visibility and accessibility, access to electrical infrastructure, proximity to the terminal, and designated EV parking areas or zones. Some respondents opted for medium-desirability locations to balance accessibility and power source proximity.

Twenty-six respondents provided information on their different levels of charger. Twenty-two of the 26 respondents had installed Level 2 chargers with the remaining four respondents only having Level 1 installed. Nine respondents had a mix of charging levels in the parking facilities. Factors affecting the number of installed charging stations included

- EV adoption in the region;
- Available electrical capacity;
- Availability of funding, grants, and incentives;
- Results of a pilot/trial of EV charging equipment in the parking facilities; and
- Results of a parking study.

## Management of Charging Infrastructure

Fourteen survey respondents do not monitor the use of their charging equipment at this stage, and ten (out of thirteen) respondents have networked chargers. This may be partially because of how charging equipment data are used—only seven respondents use their data to determine the demand for additional charging stations. The costs associated with monitoring usage and with networked chargers may limit the practice at the airports surveyed.

The airports that do monitor the use of their charging equipment collect that data for various reasons:

- Tax reporting,
- Environmental metric tracking,
- Power consumption,
- Dwell time on chargers,
- Uptime and maintenance needs, and
- Reporting for environmental or accreditation programs.

Although most respondents have not developed metrics related to the charging equipment, they noted plans to do so in the future as well as ongoing studies that could be used to develop such metrics.

## Utility Coordination and Load Management

The survey asked respondents about electrical capacity assessment, utility coordination, and load management concerns or strategies. Eighteen respondents had conducted an electrical capacity assessment before installing charging at their airports with some having plans for future assessments to ensure available capacity. Respondents who did not conduct an assessment indicated that they did not consider it necessary for their airport or were planning to conduct an electrical capacity assessment only with a significant expansion of EV charging equipment.

Sixteen respondents coordinated with local utility companies for EV charging station installations. Coordination varied from virtual meetings discussing needs to utility companies subsidizing installation costs and handling revenue collection. Some airports engaged in discussions about future expansions and available utility resources, while others coordinated for capacity expansions and infrastructure support such as solar and net metering. A key theme in the responses was coordination to ensure available capacity now and in the future. A few respondents noted that coordination was initiated as part of a study or capacity assessment.

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Although only 2 respondents currently use load management strategies, 16 respondents plan to implement such strategies to mitigate the effect of additional load from EV charging stations. Respondents were in different stages in terms of their load management needs; next steps included research, vendor discussions, load bank studies, and implementing systems to reduce charging during peak demand or load-shedding events. Some airports reported no issues but plan to coordinate with utility companies for future installations of Level 3/DCFC chargers.

## Funding and Revenue

Funding to support EV charging installations was primarily derived from airport revenue, state grants, and capital improvement budgets. One respondent used public-private partnerships in the form of a long-term lease. In this case, the private partner covers the operations and maintenance costs. In addition, one respondent stated that a third party handles the operation and maintenance of the EV charging station.

Funding for operations and maintenance was derived from parking revenue or general airport funds or is part of the operations and maintenance budget for either the airport or the parking facility. Budgeted funds may come from various sources, so respondents could not indicate a specific revenue source.

Twenty airports responding to the survey do not assess an additional fee for using the charging equipment provided in their parking facilities. The five respondents that assess a fee used a range of methods (e.g., a flat fee, fees assessed by the private provider, and fees structured to recover operating costs, including tax requirements). Certain states have adopted a per-kWh fee for public EV charging stations, which airports are required to comply with when installing their stations (Russell 2023, Jaros and Hoffer 2023). Eight of the respondents plan to assess a fee for cost recovery in the future. These fees would support operations and maintenance as the number of charging assets increases.

## Safety Considerations

Most airports (17 respondents out of 24) do not have specific safety requirements or procedures for EVs and EV charging equipment. However, airports ensure safety by following current code requirements, standards for installation, and building inspection protocols. Examples of these protocols and standards include not installing chargers under cover (i.e., the first floor of a parking garage) or within 6 feet of gas-powered vehicles.

Only a few airports (four respondents) provide specific safety training related to EVs and charging equipment for their staffs. Training methods vary, including designated safety and preventive maintenance classes, external contractor training, and collaboration with fire departments for EV fire response training.

Although some airports have coordinated with their fire departments (nine respondents), specific standards or procedures have not been fully developed. Discussions with fire departments during planning phases involve recommendations on charger location, access, and proximity to fire suppressants. Some airports are evaluating the feasibility of installing DCFCs alongside existing fuel dispensers, with on-site fire departments having standards of practices for handling EV-related fires.

The airports' responses indicate that, although discussions are ongoing, specific standards or procedures related to coordination with fire departments have not been fully developed. Some airports know the concerns associated with EV fires but have not implemented concrete measures.

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Collaboration with in-house fire departments, recommendations regarding fire extinguishers near chargers, and ongoing discussions with local fire departments were common themes among the airports' responses.

## Overview of Results by Hub Size

Airport size determines the level of funding received and can result in differing practices and needs in terms of EV charging infrastructure. The breakdown of respondents by hub size is as follows:

- 21 small-hub airports,
- 8 medium-hub airports, and
- 7 large-hub airports.

One key difference between hub sizes was the availability of EV charging in passenger parking facilities. All the large-hub airports responding to the survey provide EV charging and seven medium-hub airports also provide EV charging in their passenger parking facilities. One medium-hub and thirteen small-hub airports do not currently have EV charging infrastructure available for passengers.

Other key findings based on hub size include the following:

- In terms of level of charger, Level 2 chargers were the most common across airports of any size. Only respondents from large-hub airports indicated the presence of Level 3 chargers. Level 1 charging was reported at airports of all sizes—often to support long-term parking facilities.
- Data monitoring occurs at airports of all sizes, but large-hub airports were more likely to monitor usage and use that data to understand demand for charging in the future.
- Eighteen respondents indicated that an electrical capacity assessment had been conducted before installing EV charging stations, and six respondents reported that they did not conduct an electrical capacity assessment. In terms of hub size, respondents from seven (out of ten) small-hub airports, all six respondents from the medium-hub airports, and respondents from five (out of eight) large-hub airports have conducted an electrical capacity assessment.
- Most airports responding to the survey did not have specific safety requirements related to charging infrastructure. By hub size, only one out of ten small-hub airports had specific safety requirements in place compared with two out of six medium-hub airports and four out of eight large-hub airports.

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CHAPTER 4

# Key Insights from Airport Interviews

Assessing the airport experience with charging infrastructure offers insight about likely challenges and opportunities. In individual interviews, six airports provided information on their experiences with EV charging and how they have navigated key questions and decisions. These interviews also assessed the effect of hub size by interviewing three small-hub airports (i.e., Eastern Iowa [CID], Northwest Arkansas National [XNA], and Portland International Jetport [PWM]); one medium-hub airport (i.e., Jacksonville International Airport [JAX]), and two large-hub airports (i.e., Salt Lake City International [SLC] and San Francisco International Airport [SFO]). This chapter provides an overview of the key insights from those interviews; detailed descriptions of each case example can be found in Appendix C: Case Examples.

## The Airport Use Case

As discussed throughout this synthesis, the airport use case for charging at passenger parking facilities is relatively unique in commercial settings. Vehicles are typically parked for days rather than hours, which can mean that Level 2 chargers are underused for most of a customer's stay at the airport. Airports noted this issue when managing demand for charging; outside of valet parking, there are few options to better use the asset. Appropriate signage can help customers follow proper charging etiquette, such as only parking in the charging space if actively using the charger. Figure 1 shows signage at XNA that states the space is for EV parking only but does not require the vehicle to be charging. Providing charging in desirable locations, such as close to a terminal, can encourage EV owners to park in such locations without using the charging; signage helps to discourage this practice.

Most airports use their parking facilities to generate revenue, but this synthesis found that most airports do not assess a fee for EV charging. Partially this is due to the existing revenue generated from use of the parking facility by airport customers. Airports also noted that the cost of collecting payment may outweigh the potential revenue at this stage. CID assesses a small fee for charging because the state of Iowa has a per-kWh fee on electricity for public EV charging stations. The fee recovers the costs of paying and complying with the fee but does not generate additional revenue.

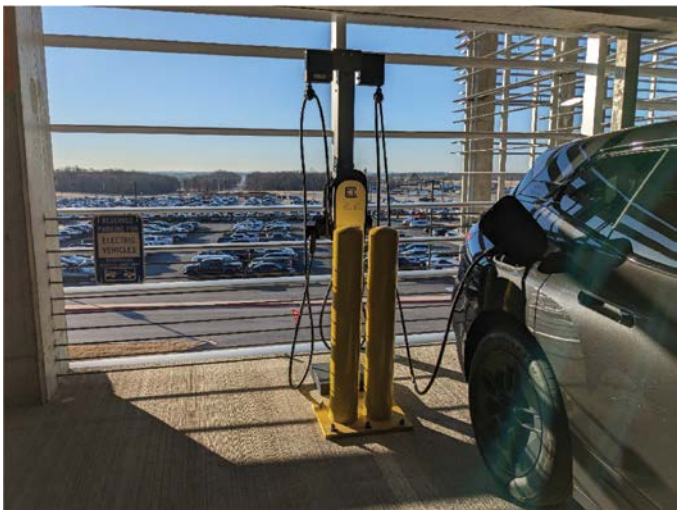
As airport EV charging assets grow and more is known about the total cost of ownership, the airports interviewed will explore possible revenue recovery methods, such as an additional fee to access charging or a per-kWh fee. The large-hub airport finance departments have shown an interest in these fees as the cost of operating and maintaining chargers increases. In addition, providing charging in non-revenue spaces such as cellphone waiting lots may prompt an airport to assess a fee, which is the case with JAX.

**SFO's Total Cost of Ownership Study**

SFO's Finance Department is analyzing the total cost of ownership for the airport's EV charging help the airport explore the need for revenue recovery and possible rate structures or agreements with their private parking managers.

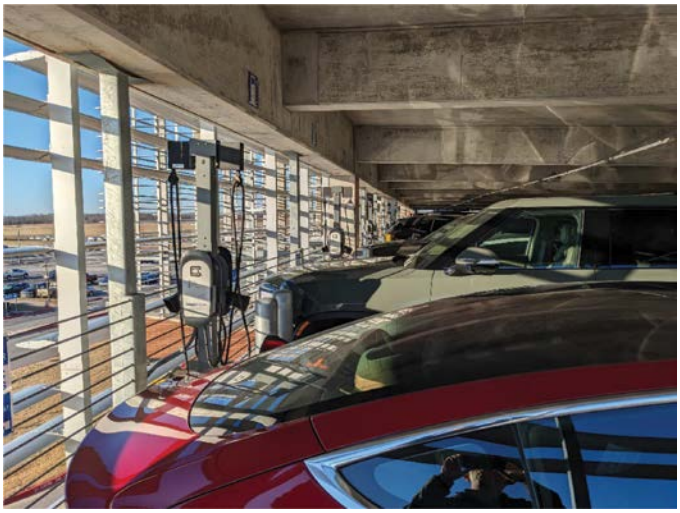


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Source: Northwest Arkansas National Airport

***Figure 1. EV parking signage at Northwest Arkansas National Airport.***



Source: Northwest Arkansas National Airport

***Figure 2. Charging equipment at Northwest Arkansas National Airport.***

The airport passenger parking use case also raises questions about the required level of charger to support customers. One airport selected charging equipment with two ports that when two vehicles are charging provides Level 1 power but when one vehicle is charging provides Level 2 charging capabilities (see Figure 2). Understanding the key decision points related to charger type was important for airports.

## Determining Appropriate Charging Equipment

Selecting charging equipment requires consideration of the airport use case, the quality of the equipment, the availability of replacement parts, and whether the airport would benefit from collecting data from the equipment.

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**Pilot Programs**

SLC staff completed a pilot to better understand EV charging needs for the airport. Pilot programs or trying out a small number of charging equipment can be an important first step in determining overall demand and the airport use case. Starting small can also help airports understand the process and work through any challenges.

**Quality of Equipment**

PWM noted the importance of robust equipment with easily replaceable parts. PWM has installed chargers in a piecemeal manner that has caused some problems. Now, the airport is trying to establish a policy and standard for the quality of equipment purchased moving forward.

All the airports discussed the tradeoffs between the levels of chargers and the specific needs at the airport. Factors to consider included the following:

- Costs of equipment versus available funds or incentives,
- Location of the asset within the airport and within the parking facility, and
- Availability of and experience with different levels of charging equipment.

Although Level 2 charging equipment fits most needs at the airport currently, futureproofing requires the consideration of all levels of charging. The high initial cost of Level 3 chargers as well as the required power to support them were cited as why the airports interviewed have not yet installed such chargers in their passenger parking facilities. In addition, funding programs or incentives may be charger-specific, which limits an airport's ability to use these funds to meet the airport's use case.

Capacity at electrical panels is more important with fast chargers but is often considered for chargers of all types. The airports interviewed either conducted an electrical capacity assessment before installing their first set of chargers or have conducted an assessment since to determine their capacity needs.

Some of the airports interviewed spoke about needing more data and information on the airport experience with each level of charger. Most airports are familiar with Level 2 but have questions about whether Level 1 chargers have worked for certain use cases or about the challenges associated with Level 3 chargers. The small-hub airports indicated a greater interest in Level 1 chargers to support their passenger parking facilities.

When discussing networked chargers and the need for data and monitoring of charging assets, responses differed across the airports. Staff at two of the small-hub airports thought that, for their purposes, data collection was unnecessary and did not warrant the cost. (One of these airports currently collects data, and the other does not.) In addition, both airports have monitored the electricity used by the charging stations to understand capacity through metering. One of the large-hub airports is not currently collecting data from their charging equipment in passenger parking facilities while they transition to a new software solution. Although this situation is acceptable in the short term, airport staff noted needing data for their fleet charging assets as well as the possible effects on load management as charging needs grow. Overall, the benefit of networked chargers that allow the airport to monitor usage and maintenance seem to be airport-specific at this stage; however, there was a widespread belief that if an airport increases its charging infrastructure, data collection will be critical.

## Software Considerations

The airports noted the importance of reviewing software contracts and management when installing networked chargers. This was especially important to the large-hub airports that had installed more charging equipment across their facilities. Software in this section refers to the payment interface and data collection systems associated with charging infrastructure. As airports increase their charging infrastructure, the need to monitor usage, uptime, and maintenance becomes more important.

SFO and SLC are exploring options to consolidate their charging equipment data under one software solution; however, this may not be practical when considering non-passenger-facing EV charging equipment. The small- and medium-hub airports did not experience the same

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issues with software management, but believe that, as their charging assets increase, the importance of clear and simple management also will increase. Regardless of data and software needs, airports should fully understand their contracts, the interoperability of the software, and how costs associated with their charging will change due to economies of scale.

Ineffective data reporting detracts from the customer experience and can cause additional work for airport staff who must check the status of the equipment physically. SFO raised questions related to airport ownership of the charging assets and what are the related benefits and costs. Could a private partner reduce the burden of EV charging and what are the tradeoffs there?

Another challenge related to networked chargers is cybersecurity; PWM has opted to keep their chargers offline, partly to avoid complex cybersecurity protocols. As standards are developed, airports should review their existing protocols to ensure the security of their charging assets and airport.

## Utility Coordination and Load Management

Utility coordination is important to avoid unnecessary setbacks, but utility timelines and forecasted need requirements differ. Airports noted forecast horizons of anywhere from 5 to 30 years; effectively planning and forecasting power demand can be difficult with the uncertainties around more energy-intensive transportation electrification, such as electric aviation.

Airports are conducting electrical capacity assessments not just before their first install but to monitor capacity needs as their charging program grows. Airports noted using consultants for such studies as well as to assist with utility coordination. Installing EV charging during construction of a new facility can enhance efficiency by ensuring that chargers are near electrical panels.

When planning for charging infrastructure, staff often must consider both the entire airport power needs as well as the "localized" needs on electrical panels. Although no airport mentioned an existing concern with available capacity or load, airports that were interviewed had made electrical upgrades or were closely monitoring their needs.

Although airport staff had mixed opinions on the benefits of data for customer demand and usage management, they believed such information was helpful in terms of load management. Networked chargers allow an airport to establish a power usage policy and to shut off charging capabilities quickly if the airport needs to reserve power. Such data are also useful in understanding the total power demand and how airports can alter charging practices to reduce spikes in use.

## Workforce Needs

Planning for, installing, managing, and maintaining EV charging requires an effort across several airport departments as well as qualified and knowledgeable staff to work on these projects. Coordination across departments requires an understanding of everyone's role and a clear management structure. Without this, day-to-day management becomes challenging as the airport staff determine who should respond to different tasks.

The planning phases also require coordination—if the charging asset is in a landside passenger parking facility, this affects the overall power requirements for the airport. Ensuring that departments communicate their needs is critical.

Several airports noted the importance of their electricians in installing and maintaining the assets. This additional job duty often is not budgeted for by these departments, but the assistance can be essential for avoiding common pitfalls and ensuring a good experience with EV charging.

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The airports also indicated that they did not have a specific budget to operate and maintain the charging equipment and were relying on existing funds from either parking revenue or operation and maintenance budgets. Without dedicated funds, it can be difficult to hire staff to meet needs. One airport noted workforce challenges in maintaining their EV charging equipment and another airport has found that their electricians have been willing to take on the additional work within their existing budget. Each airport will have unique workforce needs, but the importance of qualified and knowledgeable staff is invaluable.

## Navigating Safety Requirements

None of the airports indicated specific safety concerns about EV charging, but the airports understand the potential risks associated with EV battery fires. In navigating the safety requirements, airports are consulting local building codes and safety regulations, coordinating with their fire departments, and following relevant procedures. SFO noted specific safety regulations from their fire department; XNA's local fire department had undergone EV-specific fire training.

**Disaster Exercise: Lithium-Ion Battery Fire**

JAX is planning for a disaster exercise to focus on an EV battery fire. This exercise will test their preparedness as well as allow them to walk through the process of responding to a safety incident related to EVs.

Currently, all the interviewed airports provide some additional training to airport staff managing the stations, but all electricians are qualified and knowledgeable about the equipment. JAX has several parking garages and is planning to conduct a disaster exercise to better prepare themselves in the event of an EV battery fire.

As with utility coordination, coordinating with the local fire department and the authority having jurisdiction to determine the necessary compliance measures for codes and permitting is important. Airports refer to local codes as well as national guidance. Most airports also noted consultation with airport-specific fire departments and local fire departments for their jurisdictions, especially when charging assets were in parking garages or near other fueling infrastructure.

## Using Airport Contacts

When asked what advice airports had in terms of EV charging at their passenger facilities, several mentioned using nearby airports or airports within their network to ask questions, understand lessons learned, and avoid similar challenges. Each airport will face different challenges but communicating and coordinating with airports of similar size and location can provide an opportunity for sharing knowledge. The survey results indicate that not all airports have installed charging at their facilities yet, so the importance of information and experience sharing remains relevant, regardless of what stage an airport is at with their EV charging infrastructure.

## Conclusions

The interviews reflected common challenges across airports of any size; however, large-hub airports typically have a greater amount of charging infrastructure, which has raised specific concerns. The similarities and differences in the airport experience with EV charging in passenger parking facilities according to hub size are summarized as follows:

- Decisions on when, where, and how much charging infrastructure to install often rely on common factors, regardless of airport size. Customer demand, expected demand, providing an additional amenity, and local parking ordinances can all drive the installation of EV charging stations.

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- The level of charger is often linked to the parking type, rather than the hub size. Parking at airports is measured in days, while charging is measured in hours, which leads to inherent tensions when choosing the type of charger to install. Most airports have installed Level 2 chargers, but small-hub airports were more likely to prefer Level 1 chargers to meet their needs. In terms of passenger parking, the airports interviewed did not yet see a use case for Level 3 charging stations.
- One of the reasons airports choose Level 2 chargers is because such chargers can be networked. This provides airports with easier monitoring and allows a payment system to be integrated. However, networked chargers are an additional cost and airports have not seen economies of scale as their charging assets increase. Choosing a software solution or partner requires an understanding of those costs and interoperability with other systems. This challenge was discussed by large-hub airports, but small- and medium-hub airports have not had the same issue.
- Utility coordination and load management were greater concerns for large-hub airports, often because of the increased amount of charging infrastructure at the airport. The small- and medium-hub airports have fewer than 20 total charging stations at their airports and so have not had issues with load management. For airports without any charging infrastructure, an electrical capacity assessment can help to clarify what airport parking facilities can support as well as any required infrastructure upgrades.
- Safety requirements and considerations are typically local; if the local fire department or local building codes require enhanced safety procedures, the airport will follow them. For example, the SFFD has stricter requirements for sprinkler systems in parking garages with EV charging and/or parking than the national standards.

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CHAPTER 5

# Summary of Findings

As more U.S. vehicle owners consider, and switch to, EVs, the importance of providing access to charging at airports is growing. Many airports have noted the trends and installed EV charging capacity in at least one of their parking facilities. This synthesis effort found that at least 93 out of a total of 145 (64%) hub airports in the United States provide access to charging for their customers. Installing charging has provided both an amenity for customers and helped airports work toward their environmental and sustainability goals.

The literature review uncovered information related to the different charging equipment, including charger levels and networked chargers as well as existing research on energy monitoring and optimization. However, most of the literature did not provide information specific to the airport use case, highlighting a gap in information available to airports. Despite this gap, airports are navigating the process of providing EV charging stations; these EV charging stations enable customers to fully charge or top up miles at the airport and can help ease concerns related to range anxiety.

Although more and more airports are installing chargers and developing experience with the process, challenges remain. The following are key questions for airports looking to install or increase their EV charging infrastructure:

- What type of charger? Whether to choose Level 1, 2, or 3 chargers and whether to use networked or non-networked chargers? The unique needs of airport customers can drive those decisions. Currently, most charging for passengers is Level 2.
- Where to locate chargers? Electrical capacity, customer preference, and safety affect these decisions.
- What are the total energy needs? Understanding and managing energy needs at the airport is a small piece of the charging puzzle at this stage, but airports believe that the importance will grow in the future.
- When, and how, to collect data? Data collection and monitoring demand helps to determine the energy requirements specific to charging equipment, but results in additional costs for the equipment and software.

The cost of investment in charging equipment remains an issue, but the availability of funding through public grant funding as well as utility incentives has reduced the overall burden on those installing charging infrastructure. Although funding is available, airports do support their EV charging through airport revenue, particularly parking revenue. Even if funding or incentives are used, the airport must still consider highly variable maintenance costs. Airports are still assessing how to factor in these costs to both capital improvement and operations and maintenance budgets; typically, airport parking facilities have not provided refueling options.

Given that airports have not typically provided EV charging, understanding safety requirements is also a new area. EVs, particularly EV batteries, present different safety challenges than



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the typical ICE vehicles because of the use of lithium-ion batteries. Although rare, EV battery fires present additional hazards to parking facilities in that such fires burn longer and can release toxic chemicals. Airports are following national and local codes and standards regarding EVs, but these codes and standards are still developing. Understanding the different safety parameters (e.g., sprinkler system requirements or setbacks from other vehicles) is critical as airports install more charging. Coordination with fire departments and code compliance will be key to safe access to EV charging in the future, especially if airports provide charging near other refueling stations.

Although challenges remain, airports report positive experiences with their EV charging. Providing access to charging has improved customer satisfaction and helped advance environmental goals. Engaging staff and interested parties in the process is also key to the success of EV charging. Even though workforce challenges exist, airports have used their in-house electricians to install and maintain the assets. Because these efforts are often in addition to their normal job duties, buy-in from those departments is needed. Communication between leadership, finance, environmental/sustainability offices, and facility and parking management was cited as important to continued success. Reaching out to experienced and knowledgeable contacts (e.g., other airports, charging providers, or utility representatives) can help airports avoid common pitfalls and understand potential challenges. Overall, airports are building capacity to support EV charging needs, but as demands increase, airports will need more data and information to forecast into the future.

This synthesis provides an overview of the current experience with EV charging at airport passenger parking facilities, but further research is required to help airports fully understand their needs and navigate key decisions. Future research could consider the following:

- **Comprehensive research of and guidance on EV charging equipment, including airport charging use cases for different parking facilities, tradeoffs associated with the different levels of chargers, when to select networked or non-networked chargers, the availability of charging equipment, and how this charging equipment applies to airport parking facilities.** The guidance resulting from the research could address nuances associated with private-partner involvement (e.g., structuring of lease agreements or management contracts). A decision support tool could help airports answer critical questions in various stages of EV planning, installation, and management.
- **A guide to clarify airport needs and how charging equipment can be managed effectively.** An overview of the different entities and/or departments involved as well as key personnel to include in plans and decision-making would help establish policies and procedures for management. Such entities could include environmental or sustainability offices, facilities teams, and parking managers. Guidance could include the different data needs, use of that data, and the staff responsible for collecting and managing that data.
- **Workforce analysis to understand key trends related to EVs and EV charging equipment as it relates to transportation.** Airports noted workforce challenges, and the EV industry needs more technicians and training to support re-skilling and upskilling. An overview of workforce constraints and how transportation agencies are managing those constraints could prove useful as airports build their electrification efforts. This effort could support other transportation agencies, such as transit providers and state DOTs managing EV fleets.
- **Research on the availability of EV charging at third-party or off-site parking facilities to understand the use case, management, and revenue considerations.** Airports do not typically provide fueling infrastructure for airport customers. To avoid costs and potential risks, third-party parking facilities could provide this amenity. Research could review the availability of EV charging at these types of facilities and the revenue or business model associated with the charging infrastructure.

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- **Research on safety and security issues related to electrification and EV charging infrastructure at airports.** Research could consider both landside and airside charging infrastructure and the different entities that need to be coordinated with as well as how to navigate different standards. Cybersecurity concerns about networked chargers could also be addressed. An overview of the different training options, how applicable these are to the airport environment, and case examples that highlight practices or standards that have surpassed the national minimums would also be useful.

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# Terms and Abbreviations

| | |
|---|---|
| AIP | Airport Improvement Program |
| BEV | Battery electric vehicle |
| BIL | Bipartisan Infrastructure Law |
| CHAdeMO | "CHArge de MOve," which is equivalent to "charge for moving" and is a pun on "O cha demo ikaga desuka." in Japanese, meaning "Let's have a cup of tea while charging." |
| CID | Eastern Iowa Airport |
| CCS | Combined Charging System: a standard for charging electric vehicles that can use either Combo 1 (CCS1) or Combo 2 (CCS2) to provide power up to 350kWs. Recent updates have seen 400kW CCS chargers be deployed as well. |
| CCS 1 | See CCS |
| CCS 2 | See CCS |
| DCFC | Direct Current Fast Charger |
| Demand Charge | a fee that utilities impose on customers based on the maximum amount of power they draw during a given time interval (usually 15 minutes) over a billing period. Demand charges are often applied to commercial and industrial customers, who tend to have higher peak loads than residential customers. |
| DFW | Dallas Fort Worth Airport |
| DIA | Denver International Airport |
| EV | Electric Vehicle: a plug-in hybrid or fully electric plug-in automobile |
| FAA | Facility charge assessed by the airport |
| GA | General aviation |
| GSE | Ground Support Equipment relates to service and maintenance equipment that is used to support aeronautical operations |
| ICE | Internal Combustion Engine: a car that generates power using an engine that burns fuel with air, typically gasoline. |
| IIJA | Infrastructure Investment and Jobs Act: a U.S. law enacted in 2021 that provided $1.2 trillion for transportation and infrastructure over 4 years. |
| JAX | Jacksonville International Airport |
| kW | kilowatt |
| kWh | kilowatt hour |
| LAX | Los Angeles International Airport |
| LCFS | Low Carbon Fuel Standard |
| LEED | Leadership in Energy and Environmental Design |
| Level 1 Charging Station | A 1kW EV charging station, which uses a standard 120 V outlet. It typically provides 2–5 miles of range per hour of charging. |



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| | |
|---|---|
| Level 2 Charging Station | A 7–19kW EV charging station, which uses between 208–240 V and provides 10–20 miles of range per hour of charging. |
| Level 3 Charging Station | A charging station that ranges from 50–350 kW and has typically had a manufacturer-specific charging connector. Also referred to as a "DC fast charger" or DCFC. Many U.S. manufacturers have now adopted the North American Charging Standard. Level 3 and DCFC are both used within this report. It typically provides between 180–240 miles per hour of charging. |
| Metering | The process of measuring and recording the amount of electricity consumed by a residence, business, or device. The device is often owned by the utility, but devices can be used to measure usage on a specific panel to understand the effect of EV charging. |
| MUTCD | Manual on Uniform Traffic Control Devices |
| NACS | North American Charging Standard. An electric vehicle connector system developed by Tesla. Being standardized as SAE J3400. It can support both AC charging and DC fast-charging. |
| NEC | National Electrical Code |
| Net Metering | A billing mechanism that credits energy system owners for the electricity they add to the grid. This allows them to use the electricity at any time, rather than just when it is generated. This agreement is common with renewable energy sources (e.g., wind and solar). |
| Networked Charger | Connected remotely to a larger network and part of an infrastructure system of connected chargers. Networked chargers use an internet connection while non-networked chargers are not online and cannot be accessed remotely. |
| NEVI | National Electric Vehicle Infrastructure |
| NFPA | National Fire Protection Association |
| PHEV | Plug-in hybrid electric vehicle |
| PWM | Portland International Jetport |
| Range anxiety | Concern by a person driving an electric car that the battery will run out of power before the destination or a suitable charging point is reached. |
| SFFD | San Francisco Fire Department |
| SFO | San Francisco International Airport |
| SLC | Salt Lake City International Airport |
| Software | Broadly defined as programs and other operating information used by a computer. In this report, software typically refers to the management application that allows for charging operations, billing, and energy management. |
| Submetering | Offers the ability to monitor energy usage for individual pieces of equipment, departments, or tenants, and is often used in multi-unit properties to determine individual utility usage. |
| TNC | Transportation network company |
| UL | Underwriters Laboratories |
| VALE | Voluntary Airport Low Emissions |
| XNA | Northwest Arkansas National Airport |
| ZEV | Zero-Emissions Vehicle |
| ZiGGY | EV Safe Charge's mobile electric charging station. ZiGGY is a charging robot that "meets" drivers at their spaces rather than requiring EV owners to park next to physical charging infrastructure. |

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APPENDIX A

# Survey Questionnaire and Interview Framework

Appendix A consists of the survey sent to 123 hub airports in the United States through Qualtrics and the Interview Framework used as a baseline for conducting the case example interviews with six airports.

## Survey Questionnaire

**[OPENING MESSAGE, 1st SCREEN]**

Welcome! You are receiving this survey because you have been identified as a contact with knowledge and experience related to electric vehicle (EV) charging at airport parking facilities. This study is tasked by the Transportation Research Board's Airport Cooperative Research Program. Researchers with the Texas A&M Transportation Institute are conducting this study.

Thank you for taking time out of your busy day to help us! This research will explore factors associated with installing, operating, and maintaining EV charging infrastructure as well as the potential electrical capacity, load management, and safety considerations. This online survey should take no longer than 30 minutes to complete.

Your participation is voluntary and confidential. You don't have to participate in the study and if you do participate, you can stop at any time. The researchers' survey program will have access to the participant's IP address, but it will remain confidential. Any reports generated from this research will use aggregated, not individual, personally identifiable data. Researchers will store all research records securely and will limit access to participating researchers. Your name or record of participation will be kept on a secure, encrypted drive accessible to only to researchers on this study and not be published in any way.

The final research report is envisioned to contain case examples of selected airports but will not include the name of any individual person. No case example, however, will be published without the permission of the airport.

The survey asks for some basic point of contact information so we can follow up, but only the Principal Investigator and research study personnel will access this information.
The survey webpage will remain open until January 31, 2024. Thank you in advance for your participation.

For additional information, please contact (Assistant Research Scientist at TTI and Principal Investigator for this project).



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**(((ACCEPT TERMS AND CONTINUE))) – takes survey participant to second screen.**
**(((DECLINE))) – takes survey participant to screen thanking them for their consideration.**

**[GENERAL QUESTIONS, 2ND SCREEN]**

1. What best describes your role? [CHECK BOX STYLE]
    a. Airport manager
    b. Facilities manager
    c. Parking manager
    d. Other, please specify: [BLANK FOR RESPONSE]
2. Please provide the three-letter identifier for your airport (or the airport of interest) [BLANK FOR RESPONSE]
3. What best describes the ownership of the airport?
    a. City
    b. County
    c. Airport Authority
    d. Other, please specify: [BLANK FOR RESPONSE]
4. Who manages the passenger parking facilities at your airport?
    a. Airport
    b. Private entity
    c. Other, please explain: [BLANK FOR RESPONSE]
5. How many on-site parking facilities do you have at your airport? Please describe the nature of these facilities (garages/lots) [BLANK FOR RESPONSE]
6. Do you currently have EV charging stations within your passenger parking facilities? Y/N
    a. [DISPLAY IF RESPONSE TO Q6 is N] Do you have plans to in the future?
    b. [DISPLAY IF RESPONSE TO Q6 is N] Are there any barriers that have prevented you from installing charging stations?

*Following questions will display if answer to Q6 is Yes.*

**[WHY AIRPORTS ARE INSTALLING CHARGING STATIONS, 3RD SCREEN]**

7. What drove your decision to install charging stations within parking facilities? Select all that apply.
    a. Customer request
    b. Expected demand
    c. As an amenity
    d. Other, please specify: [BLANK FOR RESPONSE]
8. In which of your on-site parking facilities do you have EV charging stations? [BLANK FOR RESPONSE]

**[QUANTITY, LEVEL, AND LOCATION OF CHARGING STATIONS, 4TH SCREEN]**

9. What types of charging equipment/stations do you have in your passenger parking facilities? Select all that apply.
    a. Level 1, Level 2, Level 3.
        i. How many of each type? [BLANK FOR RESPONSE]
        ii. What connectors do the charging stations support? [BLANK FOR RESPONSE]

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1. Images of each with brief description (select all that apply)
2. Please provide any additional information in the space provided [BLANK FOR RESPONSE]

iii. How did you determine the number of charging stations to install?
1. [BLANK FOR RESPONSE]

iv. Where are these located within your parking facility?
1. Closest to the terminal
2. In proximity to elevators or other access/egress point
3. Designated EV parking areas
4. Close to departure zones
5. Close to arrival zones
6. Other, please specify: [BLANK FOR RESPONSE]

**[MONITORING EV CHARGING USAGE, 5TH SCREEN]**

10. Do you monitor usage of your charging stations? Y/N
    a. [DISPLAY IF RESPONSE TO Q10 is Y] Are your charging stations networked or non-networked? Y/N
    b. [DISPLAY IF RESPONSE TO Q10 is Y] How is that data and information utilized? [BLANK FOR RESPONSE]
    c. [DISPLAY IF RESPONSE TO Q10 is Y] Do you use the data to determine when to add additional charging stations? Y/N
    d. [DISPLAY IF RESPONSE TO Q10 is Y] Have you created benchmarks, measures, or metrics to assess demand and assist with charging station planning? [BLANK FOR RESPONSE]

**[FUNDING OF EV CHARGING INFRASTRUCTURE, 6TH SCREEN]**

11. What funding options do you use for the purchase and installation of charging stations at passenger parking facilities?
    a. Federal grant
    b. State grant
    c. Local or regional grant
    d. Public-private partnership
    e. Airport revenue
    f. Other, please specify: [BLANK FOR RESPONSE]
        i. [DISPLAY IF RESPONSE TO Q11 INCLUDES A, B, C, OR D] If grant funding is used, what is used to provide the matching funds? [BLANK FOR RESPONSE]
        ii. [DISPLAY IF RESPONSE TO Q11 INCLUDES D] How is the public private partnership structured? In terms of revenue sharing or fee for use of space. [BLANK FOR RESPONSE]
            1. What are the airports responsibilities in terms of operations and maintenance? [BLANK FOR RESPONSE]
12. Do you charge a fee for the use of EV charging stations? *Select yes if a private charging operator assesses the fee.* Y/N
    a. [DISPLAY IF RESPONSE TO Q12 is Y] Is the fee assessed by the airport or a private partner?
        i. Airport

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   ii. Private Partner
   iii. Other, please specify: [BLANK FOR RESPONSE]
b. [DISPLAY IF RESPONSE TO Q12a INCLUDES i] How was the rate structure designed/determined? [BLANK FOR RESPONSE]
   i. Does the fee cover the costs of operation and maintenance? Y/N [BLANK FOR RESPONSE]
c. [DISPLAY IF RESPONSE TO Q12a INCLUDES ii] How much is the fee for charging?
d. [DISPLAY IF RESPONSE TO Q12 is N] Do you plan to charge a fee to help with cost recovery in the future? Y/N
e. [DISPLAY IF RESPONSE TO Q12 is N] Where is funding for operations and maintenance derived from?

**[PLANNING FOR ELECTRICAL CAPACITY, 7TH SCREEN]**

13. Did you conduct an electrical capacity assessment before installing charging stations? Y/N
    a. [DISPLAY IF RESPONSE TO Q13 is N] Do you plan to in the future?
14. Have you contacted/coordinated with your local utility with regard to EV charging stations? Y/N [BLANK FOR RESPONSE]
15. Does your airport use, or plan to use, any load management strategies to mitigate the impacts of additional load from EV charging? Y/Plan to in the future/N
    a. Please provide any additional context or details in the space provided. [BLANK FOR RESPONSE]

**[LIFE SAFETY, 8TH SCREEN]**

16. Does your airport have specific safety requirements or procedures in place for EVs and EV charging equipment? Y/N, please describe: [BLANK FOR RESPONSE]
17. Do you have specific safety training for your staff that relates to EVs and EV charging equipment? Y/N, please describe: [BLANK FOR RESPONSE]
18. Have you coordinated with your fire department/EMS to determine the need for additional standards or procedures? Y/N/Plan to do so in the future.
    a. [DISPLAY IF RESPONSE TO 18 is Y] Could you provide further details on those standards or procedures? [BLANK FOR RESPONSE]

**[CLOSING QUESTIONS, SCREEN]**

19. May the research team contact you in the future for additional information about your airport's experience with EV charging at parking facilities? Y/N
20. [IF YES TO QUESTION 19]: Please provide your full name and email address so we can contact you. [BLANK FOR RESPONSE.]

Thank you for taking our survey! Your response has been recorded.

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## Interview Framework

The following framework was used as baseline to develop specific questions for each airport interviewed. Before the interview, researchers used the survey response to update and edit the questions to better reflect the airport's existing experience with EV charging infrastructure.

*General EV Charging Questions*

1. What prompted you to install EV charging stations/trial X charging solution?
2. Can you speak to your experience with EV charging stations in your parking facilities?
    a. What do you see as opportunities?
    b. What were the challenges you face/have faced?
3. How have you determined the appropriate mix of charger type for your airport?
4. What do you wish you had known before installing charging stations in your parking facilities?
5. Do you have any advice for other airport managers relating to EV charging at parking facilities?

*Funding and Revenue*

6. What are the main funding sources you have leveraged for EV charging equipment, upgrades, and installation?
7. With what funds do you cover operational and maintenance costs?
8. How did you determine the cost per charge/rate structure or why did you decide to offer "free" EV charging?
9. Can you speak to your experience with public-private partnerships for EV charging?
10. What are the benefits of partnering with a private charging provider? What are some of the challenges?

*Electrical Capacity and Load Management*

11. How many networked chargers does your airport have? How many non-networked?
12. How do you use data on charging?
13. Have you coordinated with your utility on capacity and needs relating to EV charging in your parking facilities?
14. Do you have any advice for other airports/airport managers on data collection or use of data?
15. Have you experienced any challenges with utility coordination, load management, or demand-related charges?

*Safety Issues*

16. What safety issues have you encountered with EV charging stations?
17. Have you adopted additional standards or protocols for your parking facilities?
18. Do you have specific training related to EV charging station safety issues?

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APPENDIX B

# Documentation of Survey Results and Analysis

This appendix presents the analysis of survey responses and highlights the status of EV charging at airport parking facilities. The survey was distributed via an online platform to 123 hub airports. The survey was initially distributed on December 05, 2023, and was open until January 31, 2024. A total of 37 responses were received from the representatives of 36 airports (2 responses were received from 1 airport). In terms of hub size, 21 small-, 8 medium-, and 7 large-hub airports responded to the survey. Figure B-1 shows the map of airport locations that responded to the survey. A total of 35 responses were received through an online platform, and 2 responses were received by email. The 37 respondents were not required to respond to all questions in the survey. The online survey allowed respondents to skip questions if those questions were not pertinent to them based on their responses to the previous questions. Also, respondents were allowed to select multiple choices for a few questions. Therefore, the sample size of each question varies, and the summary of all responses is not equal to 100%.

Figure B-2 shows the distribution of airport owners who responded to this survey. Sixteen airports are owned by an airport authority, followed by eleven airports owned by a city. A county owns three airports, and a port authority owns two airports. One airport is state-owned, a city and county own one, and one is owned by an airport commission. Two airports that responded to the survey via email did not respond to the survey questionnaire.

The airports that responded to the survey stated that most airport parking facilities are privately managed. Figure B-3 shows the distribution of airport parking facility managers. Nineteen airports mentioned that private entities manage their parking facilities. Twelve airports mentioned that they manage the parking facilities, and three have a contractual service to manage them. One airport said that the airport and private entity manage their parking facility jointly.

## EV Charging Stations at Airport Parking Facility

Out of 36 responding airports (one airport had multiple responses), 14 do not have an EV charging station in their parking facilities as of February 2024. Out of 36 responding airports (one airport had multiple responses), 22 have installed EV charging stations in their parking facilities. Out of the 14 airports that do not have EV charging stations, 8 airports responded that they have plans to install EV charging stations in the future, 4 said they do not have any plans to install charging stations, and the remaining 2 did not respond. Small-hub airports represent 13 out of the 14 without EV charging and 1 medium-hub airport does not currently have charging infrastructure. Some of the 14 are considering options outside a parking garage, while others plan to make their parking facilities EV-ready during expansion.

The airports that do not have EV charging stations noted cost and infrastructure as the most common barriers to installing EV charging stations. Other barriers include a clear picture of



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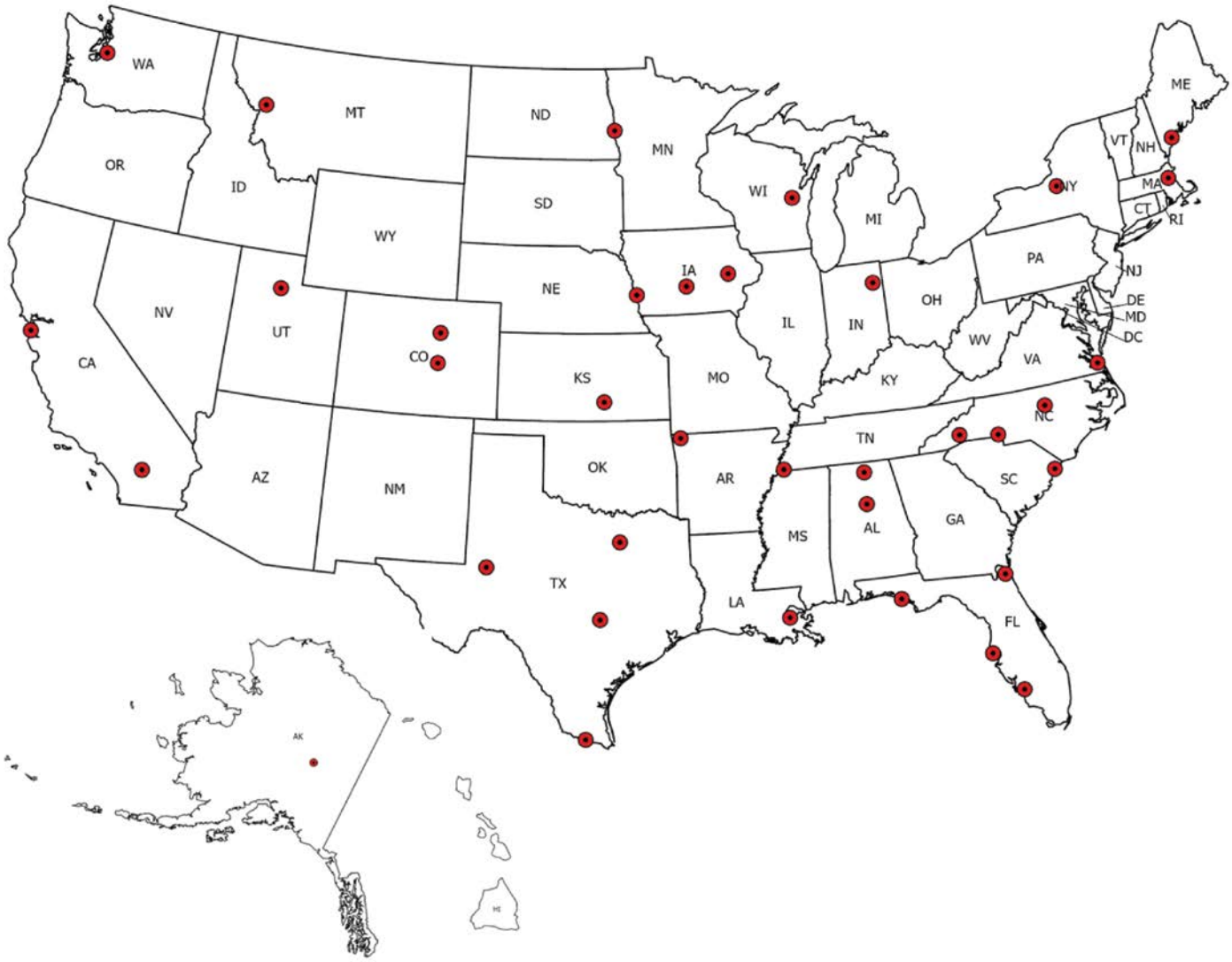



***Figure B-1. Map of airports responding to the survey.***

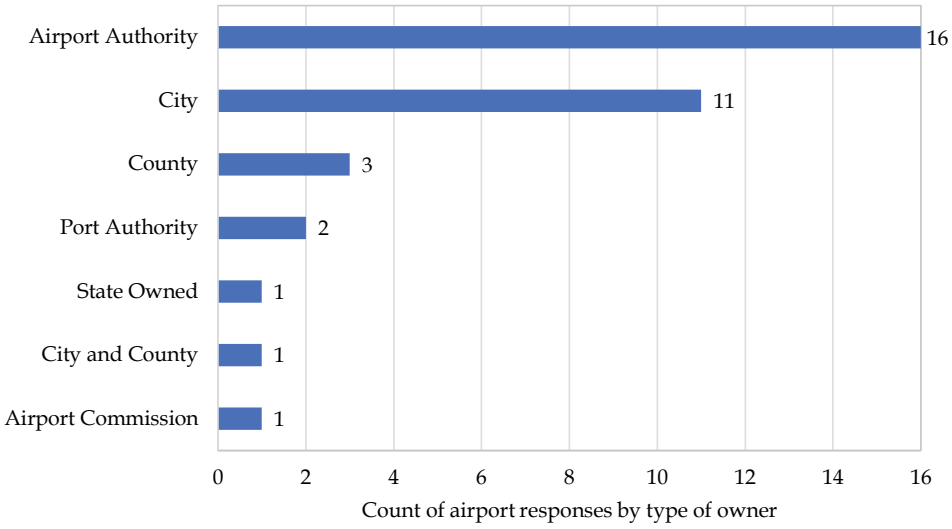



***Figure B-2. Distribution of the ownership of the airports.***

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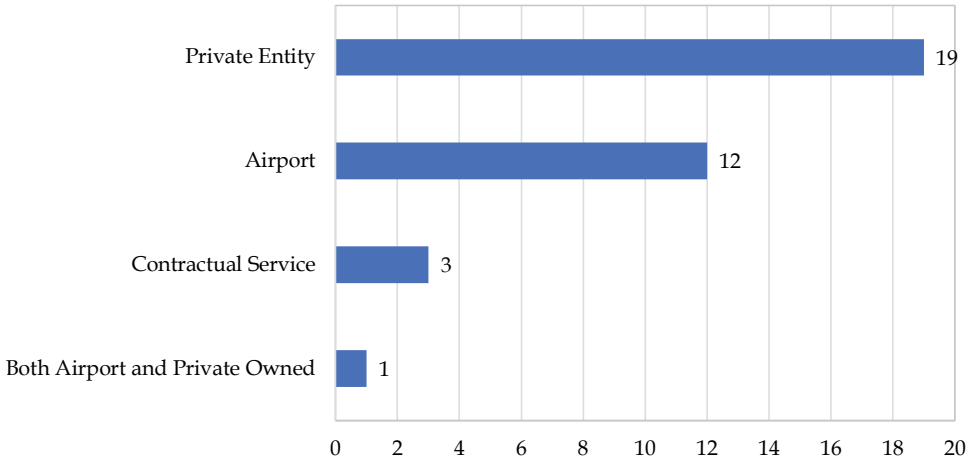



***Figure B-3. Distribution of airport parking facility managers.***

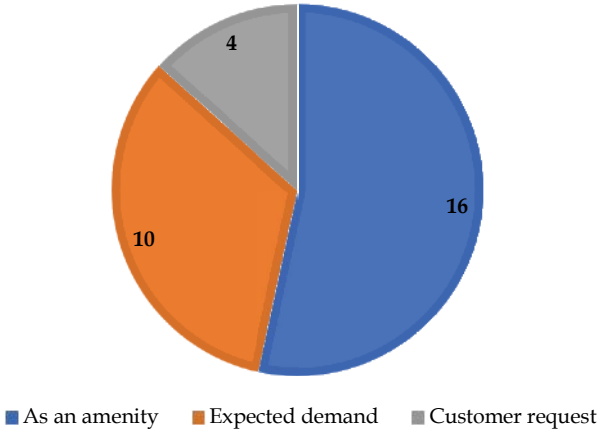



***Figure B-4. Reasons to install EV charging stations in the airport parking facilities.***

demand, management issues, maintenance contracts, on-site utility capabilities, and space. Some respondents also mentioned the challenge of vehicles being left in the charging positions for extended periods. However, a few respondents reported no barriers.

Airports installed EV charging stations in their parking facilities for various reasons. As shown in Figure B-4, out of the 30 airports responding to the question, 16 airports said it was an amenity, 10 airports said the reason was expected demand and 4 airports said customer request as the reason to install EV charging stations in their parking facilities.

In addition to the responses shown in Figure B-4, one airport responded that they added EV charging stations as an amenity and request from the rental car agencies. Another airport mentioned the reasons as expected demand and compliance with local ordinances.

## EV Charging Station Quantity and Levels

The airports have installed EV charging stations in various on-site parking facilities, including central garages, covered parking near the terminal, passenger parking decks, daily parking areas, short-term and long-term public parking lots, maintenance garages, and solar-covered valet parking areas. Some airports have EV charging stations in all their garages, while others

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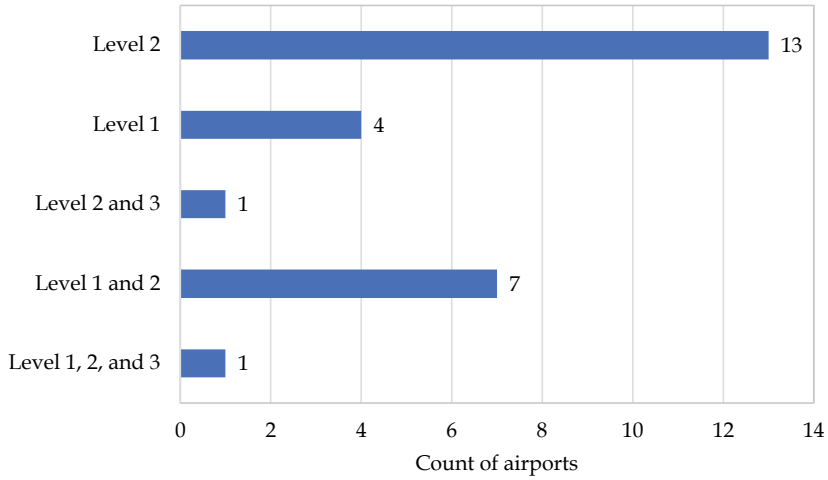



***Figure B-5. Count of airports with different charging stations.***

only have specific garages or lots. A few airports are still in the planning stages and do not have EV charging stations yet.

The reasons for choosing specific facilities for installing EV charging stations vary among the airports. Some common reasons include proximity to terminals, power availability, premium location, future demand for EV charging, and high use of customer parking. Some airports also mentioned that they chose to install EV charging stations in new garages or during the construction or expansion of existing facilities.

Some airports offer free use of EV charging stations but charge for parking, while others include the use of EV charging stations as an additional amenity with covered parking. Standard parking fees apply in some cases.

The survey results show that 13 airports have only Level 2 (240V outlet) chargers, and 4 airports have only Level 1 (Standard 120V outlet) chargers (see Figure B-5). Seven airports reported that they have Level 1 and Level 2 chargers, one reported that they have Level 2 and Level 3 chargers, and one reported that they have Level 1, Level 2, and Level 3 (480V DCFC) chargers. Figure B-6 shows the availability of different charging stations based on hub size. Only large-hub

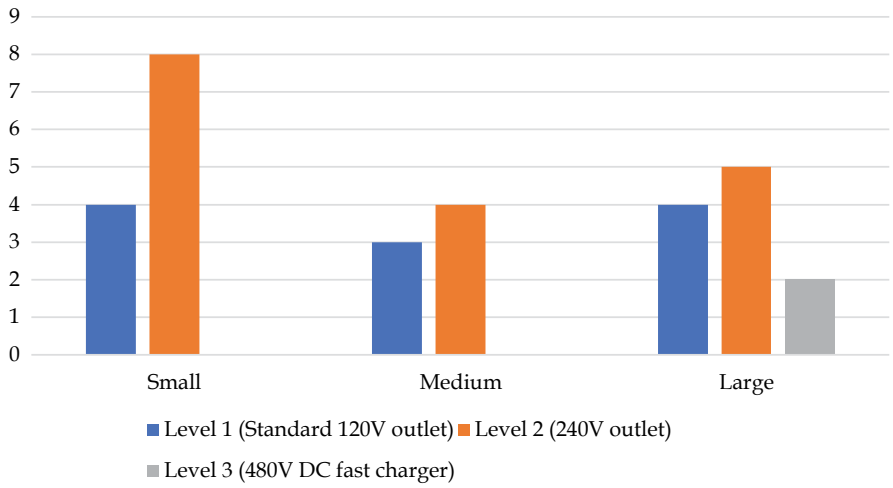



***Figure B-6. Charging station types at airports by hub size.***

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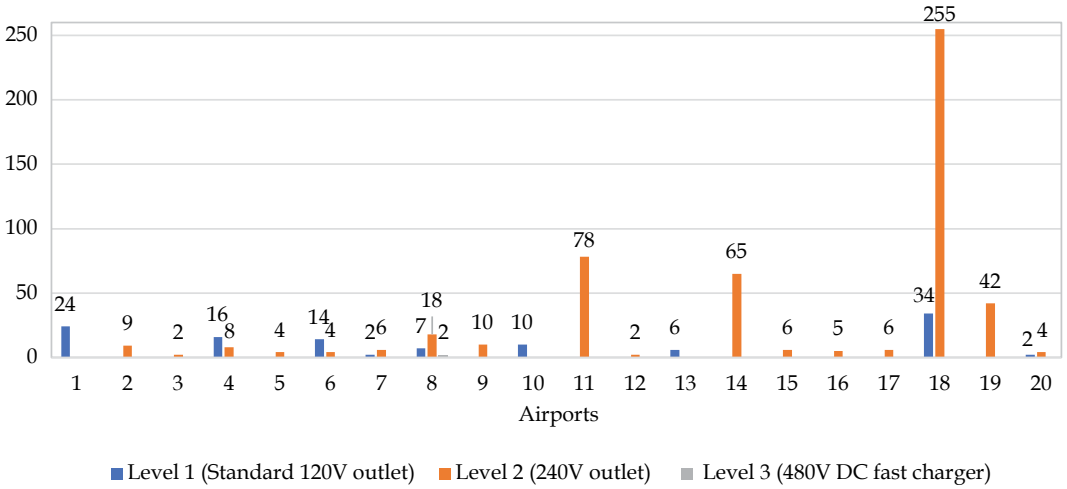



***Figure B-7. Distribution of Level 1, 2, and 3 EV chargers at airports.***

airports reported having Level 3 charging stations and Level 2 charging stations were the most common. Figure B-7 shows the distribution of Level 1, Level 2, and Level 3 chargers at airports.

Figure B-8 shows the distribution of connectors available at airport EV charging stations. Thirteen airports have Type 1 connectors, seven have Type 2 connectors, and three have CCS connectors. CHAdeMO connectors are not supported by the airports that responded to this survey as of February 2024. One of the airports mentioned plans to ensure that any vehicle can charge at any charging station, regardless of connector type. The airport is still working on the solution.

The airports indicated that the number of charging stations installed was determined according to various factors and methods, such as the following:

- Consideration of the rate of EV adoption in the region and the likely subset of that parking at the airport;
- Assessment of the availability of electrical power and the capacity of the current electrical panels;

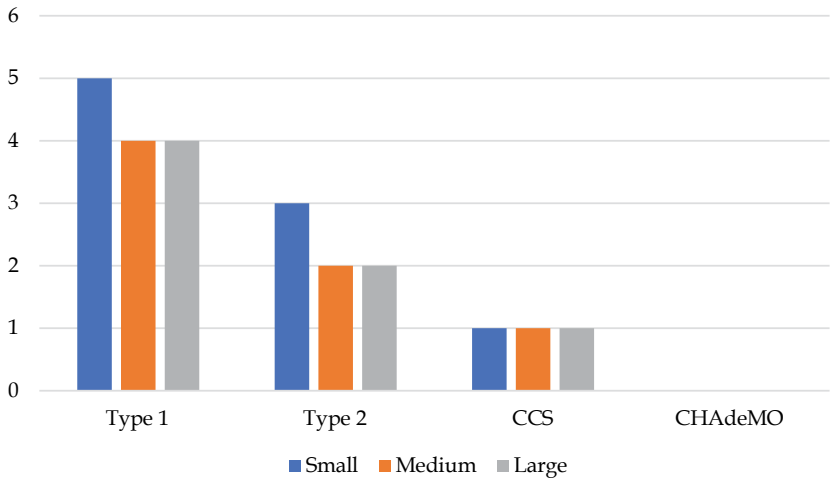



***Figure B-8. Type of connectors at airport EV charging stations by hub size.***

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- Consideration of the existing infrastructure when determining the number of charging stations to install;
- Use of grants to install charging stations;
- Consideration of available funding and consumer demand;
- Existence of constraints due to a limited budget;
- Starting with a few chargers that had monitoring capabilities to determine the number of users that used the chargers before adding additional chargers;
- Consultant recommendations;
- Installation of charging stations on a trial basis to see how much use they would receive;
- Conduct of a parking study to determine the number of charging stations; and
- Determination of the number of charging stations during a construction project.

## Locations of EV Charging Stations

The EV charging stations have been installed at various locations in the airport parking facilities, as shown in Figure B-9. Eighteen airports have designated parking areas with EV charging stations. Twelve airports have installed EV charging stations close to the terminal and seven airports have installed charging stations close to elevators or other access points. One airport has installed EV charging stations in a long-term parking facility.

The survey responses reveal that several factors determined the location of charging equipment within parking facilities. Visibility and convenience were important, with some airports believing that more visible and accessible stations would encourage greater use. The proximity to electrical panels or service, as was the existing infrastructure, was a key factor for some. Some airports installed chargers at a medium-desirability location, not too close or too far from the entrance and near the power source. Access to utility infrastructure and safety considerations (e.g., accessibility to fire trucks) also influenced the location choice.

Consultant recommendations played a role for some, while others determined locations based on feasibility and a preference toward new construction. Some advised against placing chargers close to doors or elevators to avoid tempting non-EV vehicles to park there. Minimizing installation costs was a consideration for some, as was the location of solar carport structures. Other considerations included proximity to entry doors, existing electrical service, and the access/egress location to the terminal. Some airports aimed to make it convenient for EV drivers and located the chargers in noticeable locations, frequently near the front of the garages.

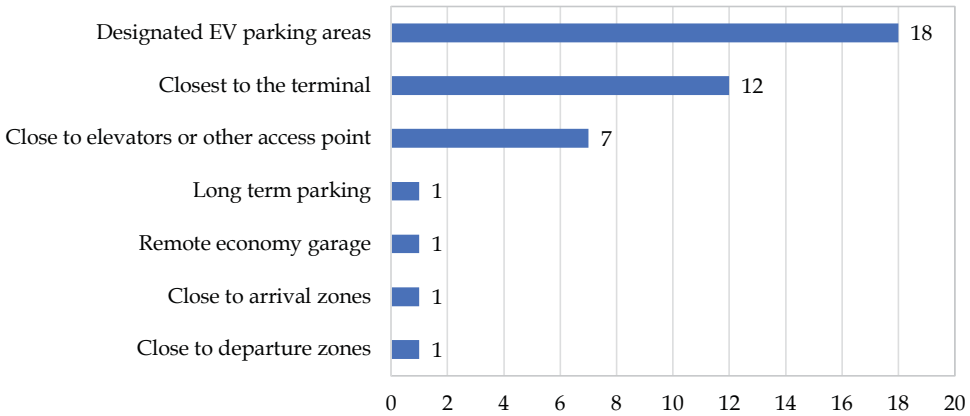



***Figure B-9. Locations of EV charging stations in the airport parking facilities.***

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## EV Charging Data Monitoring and Usage

As shown in Table B-1, 12 airports with EV charging stations monitor the usage, and the remaining 14 do not monitor the usage. Out of 13 airports, 10 airports have networked charging stations. Seven airports out of 13 responded that they use the usage data to determine the additional EV charging station needs.

Figure B-10 shows the responses to questions in Table B-1 by hub size. Five large-hub airports out of seven surveyed were monitoring data usage while three out of twenty-one small-hub airports were doing the same. Networked chargers were more common in medium- and large-hub airports; however, a similar number of airports reported using the data to determine when to add charging stations. Overall, large-hub airports are monitoring usage to a greater extent based on the responses to the survey.

In response to the survey question about creating benchmarks, measures, or metrics for assessing demand and assisting with charging station planning, airports provided varied feedback. Some are actively working on it, while others have not yet started. A few airports track usage data and some expressed interest in future metric development. One of the airports noted that they analyze the data but have not created defined metrics for looking at demand or station planning. Another airport said that ongoing energy master plans aim to address these studies.

Airports provided diverse responses regarding using data and information beyond their original purpose. Some airports are looking into data usage and determining alternate use cases. One airport employs the data for tax reporting to the Department of Revenue. Another emphasizes environmental benefits, including gas savings and reduced carbon footprint. Additionally, data is

**Table B-1. EV charging station monitoring and usage response.**

| Question | Yes | No |
|---|---|---|
| Do you monitor the usage of your charging stations? | 12 | 14 |
| Are your charging stations networked? | 10 | 3 |
| Do you use the data to determine when to add additional charging stations? | 7 | 6 |

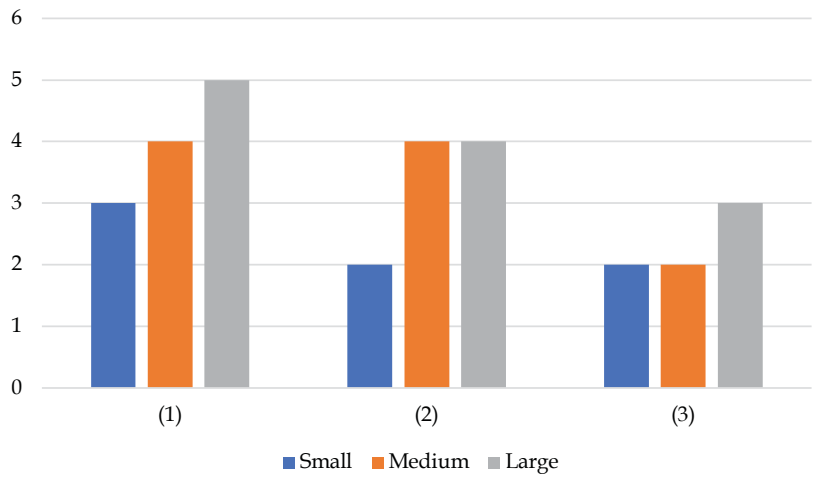



***Figure B-10. EV charging station monitoring and usage by hub size.***

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used to track power consumption and average dwell time on chargers, assess uptime and maintenance needs, and report Low Carbon Fuel Standard credits.

## Funding of EV Charging Infrastructure

Figure B-11 shows the funding options used by airports for the purchase and installation of EV charging stations at passenger parking facilities. Sixteen airports responded that they had used airport revenue as the funding source, followed by seven airports that used state grants for purchasing and installing EV charging stations. One airport used a capital improvement project budget to purchase and install EV charging stations in addition to the local or regional grant. Along with the grant funding, three airports used airport revenue, two airports mentioned airport funds, and two airports mentioned the use of capital improvement budget for matching funds. Small- and medium-hub airports were more likely to use grant funds, but airports of all sizes reported using airport revenue to support EV charging stations. One large-hub airport that used public-private partnership and federal, state, and local funding said that the public-private partnership is structured as a long-term lease. One airport mentioned that they maintain the EV charging station, and another mentioned that a third party handles the operation and maintenance of the EV charging station.

Out of 25 responses, 20 airports reported that they do not charge a fee to use the EV charging station, and 5 airports charge a fee to use the EV charging station (see Figure B-12).

Out of the five airports assessing a fee to use the EV charging station, the breakdown by hub size was one small-, one medium-, and three large-hub airports. Two airports reported that they assess the EV charging fee, and two airports reported that private partners assess the fee. One airport reported that a city-owned utility company set the EV charging fee. One of the airports where the airport assesses fees said that the fee structure is designed to recoup the operating costs and associated taxes with EV charging stations. They also mentioned that the fee covers the operation and maintenance costs of the EV charging stations. One airport reported that they do not have a submeter for the EV charging stations. However, the EV charging fee is included in the parking fee if the user parks in an EV charging station. One airport mentioned charging a flat $5 fee for the EV charging station parking spot. Out of 17 responses for this question, 8 airports reported that they intend to charge a fee to help with cost recovery in the future, and 9 airports reported they do not intend to charge a fee.

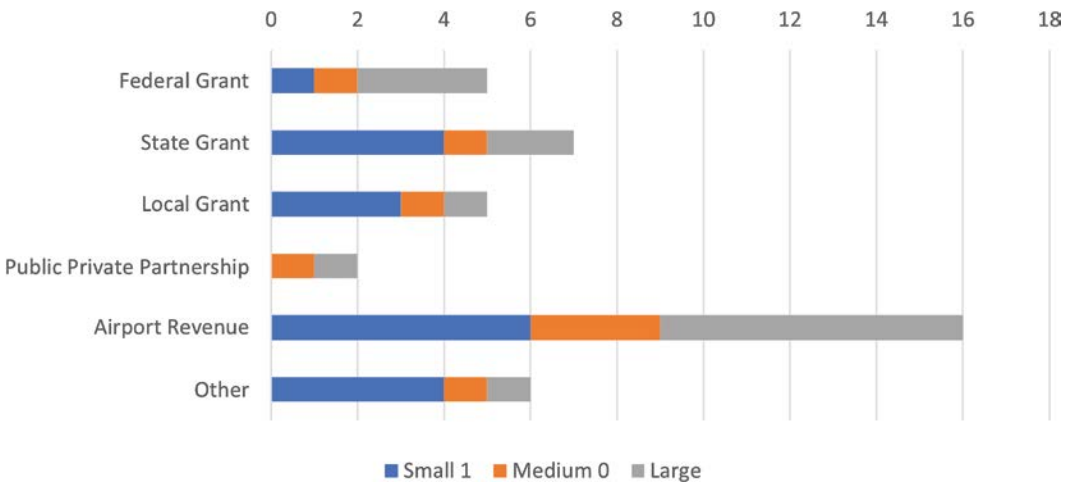



***Figure B-11. Funding options used by airports for the purchase and installation of EV charging stations.***

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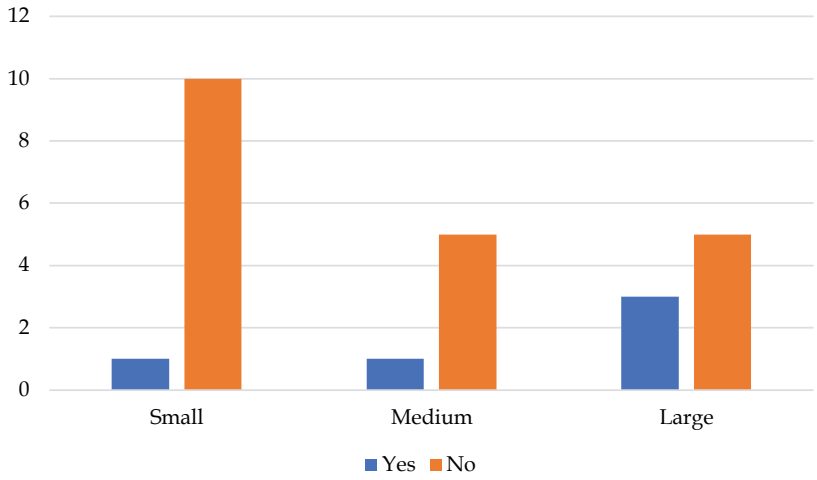



***Figure B-12. Fee to use the EV charging stations.***

Airports depend on various funding sources for the operations and maintenance of the EV charging stations. These include revenue generated from parking facilities, general airport funds, and overall parking fees. Some airports also allocate specific budgets each year for operations and maintenance. Two airports mentioned that there have been discussions and assessments about charging fees for EV chargers, but this has not been implemented yet. Additionally, third-party operators and airport revenue contribute to sustaining these services.

## Electrical Capacity Assessment and Coordination with Utility Company

Eighteen airports responded that they conducted an electrical capacity assessment before installing EV charging stations, and six airports reported that they did not conduct the electrical capacity assessment. From airports responding to this question, 70% of small-, 100% of medium-, and 71% of large-hub airports have conducted an electrical capacity assessment. Three airports said they plan to conduct an electrical capacity assessment, and two airports said they do not plan to conduct an electrical capacity assessment. One airport mentioned that they will conduct an electrical capacity assessment if they do a large-scale expansion of the EV charging stations in their parking facilities.

Sixteen airports said that they contacted or coordinated with the local utility company regarding the installation of EV charging stations. Eight airports have not contacted or coordinated with the utility company. The coordination with local utility companies regarding the EV charging stations ranged from a few virtual meetings to conducting studies to assess the need and install the EV charging stations. Airport and utility company coordination regarding the installation of EV charging stations can be summarized as follows:

- Conducted virtual meetings to discuss the airport's needs.
- The utility company installed the EV chargers free of charge through a grant. The details of the grant were not provided.
- Discussions about future expansions and available utility resources.
- The utility company will visit the airport facility to assess the need.
- The utility company subsidized the installation and network fees; however, they collected all revenue from the EV charging station usage.
- The airport designed a new garage facility for future EV charging stations and informed the utility company.

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- Conducted electrical study to assess the installation of new charging stations.
- Coordinated with the utility company to expand capacity at selected locations.
- Coordinated with the utility company for solar and net metering infrastructure to support the EV charging stations. The installation also required a new transformer.
- Purchased Level 2 EV charging stations and will receive a rebate from the utility company.

Twenty-five airports responded to the question. Currently, two airports use load management strategies to mitigate the effect of additional load from the EV charging stations. Seven airports do not use any load management strategies. Sixteen airports mentioned that they plan to use load management strategies to mitigate the effect of additional load. Four airports said they are looking at load management strategies through research, talking to different vendors, or conducting load bank studies as part of a sustainability plan. One airport said that they plan to have a load management system that would allow them to reduce EV charging during peak demand or a load-shedding event. One airport said that they do not have any issues related to load management with the existing Level 2 chargers but will coordinate with the local utility company and load management vendors because the airport wants to install Level 3 DCFCs. One airport mentioned that they had set a hard limit for charging sites with limited capacity and implemented a load management system for individual chargers to avoid exceeding the capacity. They have also conducted an EV charging capacity assessment of the airport facilities to identify areas with the most capacity to install EV charging stations.

## Safety Requirements for EVs and EV Charging Stations

Out of 24 responses, 17 airports responded that they do not have specific safety requirements or procedures for EVs and EV charging equipment (see Figure B-13). Two airports mentioned having first responders training for EVs and EV charging equipment. One airport mentioned that they have a safety procedure with their insurance company. One airport mentioned that installing all EV charging stations must follow the current code requirements. One airport mentioned that they have standards for EV charging equipment installation at the airport for power supply and transformer requirements. The same airport also requires all EV charging station installations to go through their building inspection and code enforcement group, to ensure that all applicable code laws and standards are satisfied. One airport provided examples of the safety requirements, such as no chargers can be installed under cover (i.e., the first floor of a parking garage) and EV charging stations cannot be installed within 6 feet of a gas-powered vehicle.

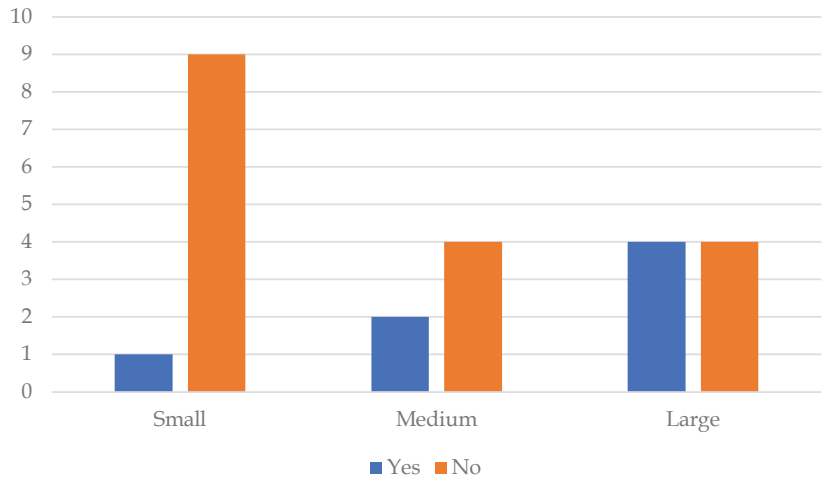



***Figure B-13. Safety requirements for EV charging stations at airports.***

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Out of 24 responses, 4 airports have specific safety training related to EVs and EV charging equipment for their staff while 20 airports do have specific safety training for EVs and EV charging equipment. One of the four airports has equipment that can fit in the garage in case of a fire. One airport asks each staff member to go through designated safety and preventive maintenance classes on EV charging stations. One airport provides training through an outside contractor, and another trains its staff through a fire department with EV fire response training.

Out of 25 responses, 9 airports said that they have coordinated with their fire departments to determine the need for additional standards or procedures related to EVs and EV charging equipment. Out of 25 responses to the question, seven airports said they have not contacted the fire department, and nine airports said they plan to contact the fire department.

In terms of airports coordinating with their fire department, or the local fire department, these discussions can happen as early as the planning phases and include recommendations on charging equipment location, access, and proximity to fire suppressants. One airport is evaluating the feasibility of installing DCFCs with existing fuel dispensers in a quick turnaround facility and are evaluating the feasibility of providing space for both. One airport's on-site fire department has standards of practices for handling EV-related fires.

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APPENDIX C

# Case Examples

## Eastern Iowa Airport

**Eastern Iowa Airport (CID)**

City: Cedar Rapids, IA
Hub Size: Small
Enplanements (2022): 611,010
Public EV Chargers: Two Level 2 (4 ports)

### Overview

Located in Cedar Rapids, Iowa, Eastern Iowa Airport (CID) is owned and operated by an airport commission while the parking facilities are privately managed. CID provides passenger parking with a short-term surface lot and a long-term surface lot. CID first installed chargers during a rehabilitation project of the parking facilities. This allowed the airport to take advantage of an ongoing construction project to install their first set of chargers. CID currently has two Level 2 charging posts, each with two ports. The posts are near each other but one serves the short-term parking lot and the other serves the long-term parking lot. The chargers are in this area to have easy access to utility/electrical infrastructure.

### Airport Charging Infrastructure

CID first installed charging stations as part of their efforts to improve environmental stewardship. Additional projects under this umbrella include solar installations. The airport also took advantage of necessary maintenance to their parking facility to install the chargers. A rehabilitation project allowed them to assess existing utility infrastructure and locate their stations in that area.

### Selecting and Installing the Charging Equipment

Before CID installed EV chargers, CID staff consulted with an engineering firm about the location of the chargers and for coordination with the utility on the necessary infrastructure. As a result of these consultations, the airport installed an additional transformer to upgrade the airport's utility infrastructure. The charging equipment does not represent a large draw in terms of power, but the transformer helps support other upgrades and power needs at the facility.

The chargers were installed as part of CID's sustainability program. Location was mostly determined by the existing utility infrastructure, but it also allowed them to co-locate ports for both the short and long-term parking lots. CID chose to install Level 2 chargers because these were the most available for purchase at the time. Two charging stations is meeting demand, but the airport will assess the need for additional chargers as the market changes and EV sales grow.

### Managing and Maintaining Charging Equipment

With two chargers, the airport can easily manage their existing assets but having a trained electrician on staff is critical. CID's electrician can conduct most of the maintenance and works well



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with the local utility. CID's IT department also plays a role by managing software issues and basic maintenance tasks that do not require an electrician. The airport did have an issue with a ground fault that was severing communication to the charger. This may have occurred when other maintenance work was happening in the area, so it is important to exercise caution around the equipment. The ground fault issue is the only major challenge that CID has faced with the EV chargers.

Networked chargers were installed which helps with managing the assets. The data provided alerts the airport to certain maintenance needs and if the asset is down. This reporting by the asset has given CID staff a better sense of the overall functionality of their charging by reducing the need to check the asset physically.

## Networking and Data Use

CID selected networked chargers but does not use that functionality much. Given the number of chargers and existing demand, CID staff do not think networked chargers are strictly necessary for their needs; however, this may change as their charging assets increase or use changes. The information is used primarily to report electricity use for Iowa's per-kWh tax on public EV charging.

Some of the advantages of networked chargers are not necessary for the CID's use case. For example, chargers that are part of a network can be easily marketed to users. However, airports are not trying to attract traffic to their charging assets, but to provide an amenity for customers. When the data is not required for load or demand management, data collection may not be needed to provide adequate charging at small-hub airports.

## Load Management Considerations and Utility Coordination

CID does not currently use any load management techniques because they have a small number of chargers. Outside of installing the transformer during their parking rehabilitation project, the airport does not coordinate with the local utility specifically about EV charging needs.

## Charging for "Charging" and Other Revenue Considerations

CID used commission funds to complete the parking project, including purchasing the EV charging equipment. The airport assesses a fee on charging to recover the costs associated with paying the per-kWh fee to the state. This fee does not cover the cost of maintenance. Operations and maintenance funding is covered through the budget for the parking facilities. The private charging provider manages the payment system that collects the fee.

## Ensuring Safety

Although CID does not have specific training or safety procedures, CID staff noted the importance of having a trained electrician on staff when working with high voltage. When purchasing EV charging equipment, maintenance will often fall to the airport or parking manager. Charging providers may offer to conduct repairs or maintenance, but it is more cost-effective for an airport to conduct its own maintenance. It is important to understand these requirements and have staff that can manage the asset.

## Lessons Learned

**1. Importance of Right-Sizing Charging Equipment**

In the short term, small-hub airports may not see the demand to support Level 2 installations. Level 2 chargers typically can provide a full charge in hours, while most airport customers

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are parking for several days. Level 1 chargers can provide enough power to support that use case but have not been used extensively in commercial settings. In addition to understanding the charging needs at the airport, it is important to gauge the proximity of additional charging infrastructure to understand demand. This can help to ensure that the assets provided on airport truly support customer need.

CID has rarely seen their Level 2 assets used at full capacity, which indicates that Level 1 may provide a more cost-effective investment in the future. Level 2 charging may work better in shorter stay facilities (e.g., the cell phone waiting lot where customers typically spend less than 2 hours).

**2. Airport electricians play a key role in supporting charging infrastructure.**

Having airport staff available who can manage and maintain charging equipment helps to (1) reduce costs by avoiding service fees from the charging provider and (2) ensure faster repairs when needed. CID noted that it would be a challenge for operations and maintenance if they did not have a qualified electrician on staff.

### Remaining Challenges

Most commercially marketed charging equipment is Level 2 or above; while these types of chargers support most commercial applications, airports often have customers parking for much longer periods than other commercial applications. Understanding the potential and the availability of Level 1 charging equipment remains a challenge. CID also noted a need to understand the airport experience with Level 1 to answer questions related to customer satisfaction and use of Level 1 versus Level 2 chargers, as well as potential challenges related to using Level 1 equipment.

## Jacksonville International Airport

### Overview

**Jacksonville International Airport (JAX)**

City: Jacksonville, FL
Hub Size: Medium
Enplanements (2022): 3,177,393
Public EV Chargers: 5 Level 2

Jacksonville International Airport (JAX) is in Duval County, Florida, about 13 miles north of downtown Jacksonville. The airport is owned by an airport authority and the parking facilities are managed by the airport. JAX has three passenger parking facilities: an hourly garage, a daily garage, and a daily surface lot. JAX has installed chargers across their parking facilities and currently have six Level 2 chargers, including one charging post at their general aviation (GA) airport. The other chargers are in each of their passenger parking facilities. These chargers are located to provide easy access to the terminal as an amenity for EV owners (except for the daily parking garage which had limited capacity on the electrical panels closest to the terminal). JAX is undergoing several changes and expansions of their parking at the airport, which will increase charger availability over the next few years.

### Airport Charging Infrastructure

Up to this point, decisions on charging infrastructure have been driven by customer demand for access to charging at the airport. JAX is rebuilding and expanding the current parking facilities. More passenger parking will be available closer to the terminal because the employee lot will be relocated further away. As part of the project, JAX will be installing additional chargers, including chargers for the employee lot. These changes will create more revenue-generating parking spaces closer to the terminal.

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## Selecting and Installing the Charging Equipment

JAX first installed chargers because of customer requests along with expected demand at the airport. The airport's current lack of parking in general has led to the expansion efforts. The installation of chargers has been a success with customers who are happy with the amenity provided in the parking facilities. Because of grant stipulations, JAX has installed Level 2 charging posts and has found that these posts meet their customer needs. Level 2 equipment is also cost-effective versus fast chargers, so future installations will also be Level 2. JAX is working with Tesla to install some of their Level 2 options because JAX staff have noticed a preference for the brand from Tesla owners. The airport is also working with several other manufacturers to install charging stations as well.

As noted, JAX installed charging across their parking facilities, including their valet parking. However, the charging in valet parking has not been well-used. This may be because employees have not promoted the service or been comfortable using the equipment, so the airport will revisit this issue with the new parking management for valet. Currently, the chargers at valet are open to employee use because they are underused by existing customers.

The first installation of chargers was intended to meet demand but also allowed the airport to understand usage patterns and needs. With future installations, JAX is selecting locations that maximize usefulness for the customer while ensuring that the power needed to support the equipment is available—sometimes this is difficult to achieve. The airport has plenty of capacity in locations that are further from the terminal, but installing chargers in such locations may not generate enough use (and therefore revenue) to support the investment. The airport is working with consultants to assess these tradeoffs and make decisions on optimal locations for different types of chargers. Location decisions are especially important, but also complex, when considering Level 3 infrastructure. JAX has considered placing Level 3 chargers in their cellphone lot, but because this is not a revenue-generating lot, the additional draw in terms of power and the management of the facility required make it difficult to justify doing this.

## Managing and Maintaining Charging Equipment

Although the charging stations have been a success, the current availability of charging does not meet demand within the facilities—partly because parking at the airport lends itself to longer stays than a typical charging session would require. Emphasizing charging etiquette remains a key component to meeting customer expectations. JAX staff have noticed that drivers are happy to release their plugs once their charging sessions have ended to allow others to use the equipment. Although this is not a long-term solution, it is reducing demand issues somewhat.

Management of the charging assets has been relatively straightforward for JAX. Most challenges stem from the lack of parking space overall. For example, it was difficult to block off the space to install the chargers—Customers would move cones, so the airport used their own vehicles to secure the spots. One tactic the airport has used when needing to install or perform maintenance is to communicate through the charging provider's app. This level of communication has ensured customers are not confused or unaware of planned outages. JAX is upgrading the parking facilities, but when the upgrade is occurring, more plug-in hybrid owners could use the charging spaces because of the access to the terminal and overall convenience.

So far, maintenance has been relatively straightforward. Any issues have been quickly resolved by the manufacturer, and replacements have been provided if necessary. JAX electricians service the equipment and have become knowledgeable about EV charging and the associated equipment. This has eased any maintenance issues with respect to the charging equipment.

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## Networking and Data Use

All the existing chargers at JAX are networked—the airport monitors usage and other available data through the dashboard. The existing chargers are from the same equipment provider, which allows for easy monitoring. JAX is not concerned about using multiple vendors as they install more chargers, but this may become a consideration as the need for charging grows. JAX staff have heard from one provider that the provider's software can cover all the different charging solutions except Tesla.

## Load Management Considerations and Utility Coordination

Networked chargers allow for better data availability, which is important for load management and coordination with the utility. Using information from charging equipment providers, the airport conducted an electrical capacity assessment before installation; this study was landside focused. With the number of chargers currently installed, JAX does not see load management as an issue for the public parking facilities. However, private partners wanting to install Level 3 chargers and rental car companies electrifying are considerations and highlight the need to coordinate with the local utility.

JAX has discussed these electrification needs with the local utility and determined the need for new power lines, transformers, and, ultimately, a second substation. The new power lines and transformers will support rental car operations, but the substation is intended to provide a backup and support the airport's overall power needs. The electrical infrastructure upgrades will allow the airport to increase the number of chargers available to airport customers without concerns over capacity. In addition, to facilitate their charging and power needs, the facilities team meets regularly with their utility to keep all relevant staff current on any changes. JAX tries to provide a forecast of at least 5 to 10 years for their power needs and completes a 20-year master plan that will highlight any large increases in usage.

## Charging for "Charging" and Other Revenue Considerations

JAX is not assessing a fee for charging in areas where they are already generating revenue; however, a fee is assessed at their GA airport and within their cellphone waiting lot. The reasoning for this split is that the cost of providing the electricity is negligible, especially considering the length of time most customers are parked at the airport. The fees in non-revenue-generating spaces allow the airport to maintain the equipment effectively.

The charging equipment was obtained with grant funding through the local metropolitan planning organization (MPO), North Florida Transportation Planning Organization. The grant covered 3 years of maintenance as well as the asset, which is part of the reason that JAX has had a great experience installing and managing their chargers. With that in mind, if their maintenance costs increase significantly, the airport may consider assessing a fee to cover such increases.

New charging equipment is being installed through airport revenue, specifically from parking revenue. JAX is then using regular operations and maintenance funds to maintain the chargers. The airport is open to exploring more grant opportunities, but currently has the funds to support chargers through airport revenue.

## Ensuring Safety

When installing EV chargers, JAX coordinates with the fire department and follows relevant building codes. The plan for the new parking garage is to have one row of chargers in a designated area away from other parking spaces. The fire department has been involved in these discussions

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and coordinated with the airport on the location. Another consideration for JAX is flooding, so the equipment must be raised to avoid issues in the event of a hurricane or other localized flooding.

The airport is conducting an airport disaster exercise this year that will use a lithium battery fire as the event. The intent with this exercise is to understand JAX's weaknesses and develop an action plan in case of a lithium battery fire in the parking facilities. The exercise will also allow different interested parties to understand their roles and safety issues relating to EVs and charging equipment.

### Lessons Learned

**1. Use the Experience of Other Airports**

JAX has had a positive experience with EV charging and staff emphasized the importance of talking with other airports and those knowledgeable about the process. Use all available resources, whether that be other airport facility teams, consultants, utilities, or charging providers. Asking questions early and often can lead the airport to the best possible options and help avoid obstacles and pitfalls.

**2. Communication and Coordination Across Interested Parties Is Critical**

As transportation electrifies, it is important to understand electricity needs across the airports. Rental car companies and Level 3 chargers present challenges for the airport but communicating with their utility and other relevant parties has reduced the overall burden. JAX also coordinates between landside and airside facilities to share knowledge and split some of the duties based on expertise.

### Remaining Challenges

Load management and available capacity at the airport remain a challenge. It is important to work with all the different interested parties to meet the electrification needs at the airport. This requires consideration of both airside and landside, including the facility, parking, and rental car company needs.

Capacity and availability of electrical infrastructure also affect charger location—this is a component of their EV charging that JAX is still working through. Siting chargers close to the terminal provides an amenity but increases the desirability of the spaces. JAX is trying to keep pace with demand but does not have enough charging equipment to meet its needs.

## Northwest Arkansas National Airport

### Overview

**Northwest Arkansas National Airport (XNA)**

City: Highfill, AR
Hub Size: Small
Enplanements (2022): 812,848
Public EV Chargers: 16 Level 1 and 8 Level 2

Northwest Arkansas National Airport (XNA) is in Benton County, Arkansas, northwest of both Fayetteville and Springdale. The airport is owned by an airport authority and the parking facilities are privately managed. XNA has four passenger parking facilities: three surface lots and one parking garage. XNA first installed chargers during the construction of their parking garage. This allowed the airport to take advantage of an ongoing construction project to install their first set of chargers. XNA has sixteen Level 1 chargers and eight Level 2 chargers. The chargers are in designated areas with easy access to electrical infrastructure and offer medium desirability for customers to decrease use by non-EV owners.

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## Airport Charging Infrastructure

XNA's decision to install charging infrastructure was driven by the construction of their new parking deck. This allowed the airport to take advantage of the project and include the requisite infrastructure in the planning stages. The EV charging equipment available at the airport has been installed over time and so the decisions around what to install and location have changed as the airport has collected more data on customer preferences and use.

## Selecting and Installing the Charging Equipment

XNA selected Level 2 charging equipment for the first installation. After that installation and a subsequent study on EV charging needs for the airport, XNA realized that most customers were parking for days at a time and a Level 1 charger would be sufficient. More recent installations have used chargers that have two ports; when one vehicle is using the charger, it acts as a Level 2 and when two vehicles are plugged in, the charger provides Level 1 charging to each vehicle. This setup meets the current needs of those parking EVs within the public parking facilities while keeping costs reasonable for the airport.

In terms of location, EV charging is focused on XNA's revenue parking, so although XNA has a cellphone lot, there are no current plans to install charging in that location. Location within the parking facility accounted for both the distance to electrical panels and infrastructure and the desirability of the space from a customer standpoint. Because the charging is offered as an amenity, it was important to provide a good location within the parking facility to encourage use. However, avoiding additional costs requires the chargers to be close enough to the existing electrical infrastructure.

The airport first installed charging equipment into new construction, subsequent installations have revealed challenges. For example, given the type of floor in the parking deck, some critical components in the concrete cannot be drilled through. Ensuring the contractors understand these limitations when working to install charging is critical to avoid delays in the project. XNA staff have found it important to communicate these issues and ensure the contractor is familiar with the facility.

XNA currently has enough charging equipment to meet customer needs, but in working on their parking facilities, the airport is placing conduit to support EV charging growth in the future.

## Networking and Data Use

XNA opted for non-networked chargers to reduce the capital investment cost and the ongoing maintenance costs. Networked chargers typically require a subscription with a software manager or a contract with the charging equipment provider. Non-networked chargers allowed the airport to purchase the equipment outright and eliminate ongoing costs outside of airport-provided maintenance. Given that they currently support a small amount of EV charging, the airport did not think it was necessary to network these for their purposes. Use of the parking spaces is monitored through their parking guidance system. Although this does not provide information on whether the vehicle is plugged in or not, it provides information on the number of EV owners using airport parking facilities.

XNA has noticed EVs parking in the space without using the charger, but current signage does not specify that an EV must be charging to park in that space. The charging equipment available is meeting customer demand, so there is no plan to alter signage or procedures.

In terms of advice or lessons learned for airports, XNA stressed the importance of understanding customer needs. As a small-hub airport, they have focused on non-networked Level 2

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chargers because this meets the needs of their region based on registered EVs, customer behavior, and airport capacity. XNA conducted an EV charging needs study to understand the complete picture when planning new installations, and understanding their customers was a key part of that effort. Understanding the number of EV owners using parking facilities, where they are parking, how long they are parking, and how many EV owners are in the region is key to effective planning and management.

## Load Management Considerations and Utility Coordination

XNA and their partners conducted the electric capacity assessment to understand the total power needs of their newly built parking deck and plan for future EV charging additions or electric capacity needs. The assessment showed that the transformer serving their parking structure more than met their needs, but they purchased a load meter to install on their electrical panels so that the airport could monitor and observe the load on their panels and ensure they had enough capacity. The airport electricians installed the load meter and have since removed it, but this is now a "tool in their toolbox" for use elsewhere across the airport.

## Charging for "Charging" and Other Revenue Considerations

XNA does not assess a fee, outside of the normal parking fee, for use of the charging stations. This decision resulted from factors such as ease of operations, costs, and wanting to offer charging as an amenity. In terms of cost and ease of operations, the power rate for the airport is relatively low, so the amount XNA would be spending on electricity did not justify the cost of a payment system for their chargers. As charging use and needs grow, the airport may consider recovering revenue to support operations. XNA has a system to reserve premium parking spots before arriving at the airport—this system could be used to recover revenue from the EV charging spaces as well.

Existing infrastructure at XNA was funded through airport revenue. Maintenance costs are also funded through airport revenue. The airport is open to grant funding if that supports their needs.

## Ensuring Safety

XNA has not encountered any safety issues, but the fire department is trained to handle EV incidents, such as fires. This training was undertaken by the fire department and was not prompted by anyone at the airport. For future charging equipment installations, the airport will coordinate with the fire department to ensure easy access and egress if an incident occurs.

In terms of additional standards and procedures, the parking manager required that owners of vehicles known to pose a fire risk park in specific areas to reduce the potential for a fire to spread. However, XNA has not had any issues to date.

## Lessons Learned

**1. Use Existing Data on Customer Needs**

XNA staff emphasized the importance of using the existing understanding of how and where customers park to help with charging equipment decisions. If customers typically park for multiple days at the airport, Level 1 or Level 2 chargers will provide sufficient power to customers' vehicles. Fast-charging may become important, but the cost of providing such service currently outweighs the expected usage and benefit for XNA.

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**2. Right Size Charging Equipment**

Understand needs in terms of both the customer and data for the airport. Level 1 chargers may be enough to support the needs of most customers, which can reduce investment costs. Networked chargers increase costs and may not be necessary if charging demand is relatively low.

### Remaining Challenges

XNA staff believe the airport is well-positioned with the current EV charging infrastructure; however, questions remain about (1) location choices for new equipment and (2) staying current with industry standards. XNA has been considering whether to disperse charging throughout their facilities or consolidate charging into one area. Because providing equipment across multiple locations would increase costs, the airport is currently favoring one EV "zone." Understanding the tradeoffs on location choice is important to right-sizing their EV charging infrastructure.

## Portland International Jetport

### Overview

**Portland International Jetport (PWM)**

City: Portland, ME
Hub Size: Small
Enplanements (2022): 982,834
Public EV Chargers: Four Level 1, Four Level 2, and Two Level 3 (Superchargers)

Portland International Jetport (PWM) is in Portland, Maine, 2 miles west of downtown Portland. The airport is owned by the city and the parking facilities are privately managed. PWM has three parking facilities: two surface lots and one garage. Charging stations are provided in their passenger parking garage as well as their maintenance garage to support airside fleet usage. The airport has eight charging ports available; four Level 1 chargers and four Level 2 chargers available for the public. The airport website also indicates two superchargers in one of their passenger parking lots. These chargers are near the terminal as well as supporting electrical infrastructure.

### Airport Charging Infrastructure

PWM initially decided to install chargers to improve customer satisfaction; across the state, the number of plug-in vehicle drivers has increased, and the airport saw the opportunity to provide charging as an amenity. Alongside offering charging as an amenity, the airport is supporting the City of Portland's policy on sustainability (which is focused on transitioning the city fleet to EVs and promoting the use of plug-in and fully electric vehicles by Portland's citizens). The airport has installed chargers primarily for passenger use but is expanding their footprint with chargers in the employee parking lot, equipment to support airport airside vehicles, and a few chargers for employees of the city.

### Selecting and Installing the Charging Equipment

Given that PWM has installed their charging equipment over time, the airport has used various manufacturers, depending on needs, costs, and availability. This has led to a mixture of Level 1 and Level 2 chargers. This approach, as noted by the Facilities Manager, has made clear that not all chargers are of equal quality. Acknowledging this, the airport is trying to establish a standard for charging infrastructure as they move forward. PWM's focus is on recognized brands that (1) can provide easy replacement parts and (2) are known to provide a solid product.

Previously, PWM purchased chargers at a reduced cost online, but staff noticed that they tend to be a lower quality product. For example, PWM electricians found a Level 2 charger online for $1,000 that was rated for outdoor use only. Understanding that lower cost can mean lower quality,

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the airport is focusing on robustness and availability of parts for when repairs are necessary. In terms of robustness, PWM staff noted that they have had issues with cords being ripped from the charger, often accidentally or due to user error. However, they have had success with some of the larger manufacturers in terms of parts availability to repair the equipment in those situations.

In terms of level or type of charger, PWM is questioning their charging equipment needs based on the current uses at the airport. Level 1 will likely meet the needs of their customers currently but may not in the future as charging equipment evolves. PWM is considering their futureproofing needs before purchasing new equipment. Futureproofing as it relates to PWM's charging is discussed further in the section on Load Management Considerations and Utility Coordination.

Another challenge with charging in the passenger parking facilities is a recent ordinance passed by the City of South Portland that establishes requirements for charging availability. In new parking facilities, 20% of available spaces must be able to support charging equipment. In meeting the requirements of the ordinance, PWM is aiming for 30% of their EV charging spaces to be Level 1 ready and then 40% of those spaces to be Level 2 ready. Providing this number of make-ready spaces introduces challenges for the airport, and PWM staff are working on how to bring their current parking garage up to standard. Their current strategy is to install charging equipment around the exterior near conduit and electrical infrastructure. Location of charging will depend on the proximity to the required supporting infrastructure.

### Managing and Maintaining Charging Equipment

Outside of managing issues relating to charger quality and robustness, the airport has not had any maintenance- or management-related concerns. Because the airport electricians are knowledgeable about the charging equipment, PWM has been able to realize some cost savings. For example, the electricians complete the installations themselves and conduct routine maintenance.

One of the key issues with maintenance is related to charger quality; access and availability of replacement parts is important to ensure an adequate life of the charging asset. PWM has had issues with locating replacement parts for charging equipment purchased from smaller manufacturers and brands that may not be nationally recognized.

### Networking and Data Use

PWM has opted for non-networked charging solutions at this stage, but this may be revisited as their charging equipment needs increase. Networking the chargers would provide data to support a rate structure or to understand the effect on electrical load. However, the airport has not seen the need for networking yet. In addition, networked chargers would require cybersecurity protocols and policies to be put in place.

### Load Management Considerations and Utility Coordination

With the existing charging equipment, the airport can manage the required capacity and easily coordinate with their utility on their needs. However, the new parking ordinance in Portland may require PWM to add capacity. Meeting the standards set by the parking ordinance is not the only reason that additional capacity may be needed; PWM is trying to futureproof their parking facilities to handle the charging needs of the future. Key questions they are grappling with include the following:

- How will technology improve?
- What are the implications for existing charging equipment?
- What will charging needs look like in the future?

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- How do changing needs affect required utility infrastructure and coordination on upgrades or added capacity?

PWM wants to avoid investing in charging equipment today that ultimately will need to be replaced in 5 years or building out charging infrastructure that cannot be supported by their existing electrical capacity, which may then lead to a complete facility overhaul. The airport also wants to avoid costly upgrades that prove unnecessary. For example, rental car companies approached the airport about electrification needs. These needs would have required significant upgrades, but the rental car companies have since altered their plans, so PWM was able to avoid a costly investment.

Despite these questions about futureproofing, overall conversations and coordination with the utility have been easy, given that providing enough load to the airport is not a concern. However, as the fleet turns over, PWM sees utility coordination and load management as potential issues to watch.

## Charging for "Charging" and Other Revenue Considerations

Charging equipment is funded through parking revenues. PWM can justify this use of funds because they provide charging as an amenity for their customers. The airport has not yet needed to use grant funding to support installations; however, the city is paying for two ports to be installed for use by city employees. There is no fee assessed on charging, but as capacity and costs increase, PWM may explore fees as a cost-recovery mechanism. Complying with the parking ordinance will require a significant investment, which may necessitate revenue recovery. In addition, the airport has noticed an increase in the cost of their charging equipment over time. Although PWM still can find cost-effective equipment, quality is an issue. Operations and maintenance of the chargers is covered through airport revenues and included in the general maintenance budget.

## Ensuring Safety

Excluding some minor incidents that led to damaged charging assets, PWM has not had any safety issues with their charging infrastructure. The airport does not provide specific training or guidance to their employees on using the charging equipment. However, PWM staff coordinate with the fire department to ensure the safety of the location as well as access and egress. Future installations may require additional consideration of safety regarding charger placement and location in parking garages as the airport attempts to meet the city's goals.

## Lessons Learned

**1. Consider charging needs with all new facilities.**

PWM's experience has highlighted the need to consider charging equipment when building a facility to avoid costly upgrades or retrofit projects. Including charging in the planning and design phases allows for both accurate cost estimates and the ability to locate these charging spaces close to the electrical infrastructure. Recent ordinances at the city level have highlighted the importance of futureproofing investments and being proactive in considering EV readiness as well as actual installations of charging equipment. New facilities offer an opportunity to factor in electrical capacity, utility coordination, and safety.

**2. Use the experience of nearby airports.**

PWM's advice to other airports with EV charging questions is to consult peer airports that may have insight because of experience with installing, managing, and maintaining EV charging

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equipment. Using such connections can help airports avoid some of the challenges and pitfalls that can be associated with selecting the right charger types, choosing the appropriate locations, and ensuring a reasonable investment in terms of cost.

### Remaining Challenges

PWM remains in an early growth phase of EV charging; the airport has not had a consistent policy regarding charger type and quality. Moving forward, the airport hopes to select chargers that meet their quality expectations for a reasonable price. Establishing an understanding of the correct quality of chargers will rely on their prior experience with charging equipment as well as connecting with neighboring airports to share lessons learned and potential opportunities. In addition, futureproofing the airport will require an understanding of the electrical capacity needs of fast-charging and potentially megawatt charging. The uncertainty surrounding the future of battery and charging technologies raises important questions about the correct investments, as well as correct level of investments, that airports need to make in the short term.

**Salt Lake City International Airport (SLC)**

City: Salt Lake City, UT
Hub Size: Large
Enplanements (2022): 12,383,843
EV Chargers: Sixty-five Level 2

## Salt Lake City International Airport

### Overview

Salt Lake City International Airport (SLC) is 4 miles west of downtown Salt Lake City, Utah. The airport is owned by the city and the parking facilities are privately managed. SLC has a parking garage, economy parking surface lots, and employee lots with EV charging equipment. SLC first installed chargers as a pilot program to test the potential challenges and determine the needs of the airport. SLC has 65 Level 2 outlets available across the airport. These chargers are located with easy access to the terminal. For example, in their garage, most of the chargers are near the entry doors.

### Airport Charging Infrastructure

SLC's first installation of charging equipment was part of a pilot project to study the use of and need for charging infrastructure; the project included SLC's employee lot, economy lot, and parking garage. The pilot program allowed SLC staff to understand EV charging to a greater extent, prove that the demand was there, and gain buy-in from the other airport stakeholders. The airport wants to encourage the use of EVs to help address air quality and environmental issues within the region. Providing easy access to charging is a critical component of the transition to EVs within transportation, but it also helps the airport meet sustainability goals as part of the Airport Carbon Accreditation program. After the success of the pilot program, SLC has continued to install charging equipment to meet expected demand and support sustainability goals.

### Selecting and Installing the Charging Equipment

SLC has used a phased approach in determining where and what to install across their parking facilities; they assess demand, space, and electrical infrastructure considerations every year to determine where to install equipment next. Over time, the airport's understanding of location requirements as well as the type of charging equipment to install has developed and changed. Initially, location decisions were based on charging as an amenity and wanting to incentivize EV usage throughout the region. Moving forward, SLC will be concentrating charging equipment in one area of each parking facility. Should the airport decide to assess a fee for charging in the future, this will enable them to base it on entry into the "EV Charging" section or zone. In addition, consolidation

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can reduce the burden on available electrical infrastructure by providing one area to upgrade rather than multiple.

One challenge that SLC has encountered when planning their charging installations is their stakeholder needs. When they were first planning for the new parking structure, the rental car companies did not want to include EV charging infrastructure. Now, many rental car companies have set targets and are incorporating EVs into their fleets. The rental car companies have space in the public parking garage, so SLC has provided infrastructure for Level 2 chargers for the rental car companies. This has highlighted the difficulty of forecasting needs and demands as the market changes rapidly.

SLC has installed Level 2 chargers in their public parking facilities and does not plan to provide Level 3 chargers to the public. They have installed Level 3 chargers to support their airside operations. The airport does installation, operations, and maintenance work in-house with the assistance of their electricians.

## Managing and Maintaining Charging Equipment

The nature of parking needs at an airport complicate management decisions. SLC is emphasizing charging etiquette and has updated their signage to "EV Parking Only While Charging" rather than "EV Parking," but passengers tend to park in airport facilities for longer periods than required to fully charge a vehicle. Valet parking is one option to better use their assets, but this adds expense.

Despite issues with managing use of their charging, installation and maintenance has been a success in large part due to SLC's electricians and facilities team. SLC's electricians have been willing to take on the additional work required to maintain the charging assets and have even hired an additional employee, with their own budget, to support their charging equipment. That position is to be focused on checking all the charging equipment, both airside and landside, throughout the day. The ability to conduct this work in-house has been critical to upkeep and has allowed SLC staff to explore different software solutions as their needs evolve.

## Networking and Data Use

The EV charging space has evolved a great deal in recent years and as more solutions enter the market, it is important to consider interoperability. SLC is moving toward one software solution that works for all their charging equipment, especially to support their parking garage. Using one software solution allows SLC to have a better picture of their overall demand and enables them to load manage where necessary and/or possible. In terms of public parking, the airport sees their role as providing enough charge for the vehicle to make their next trip, but not necessarily fully charge. This allows them to shift power from charging if needed to smooth peaks across the day.

SLC initially installed networked chargers and used the data for maintenance, but as the cost of the service contract increased, the airport chose to turn off that capability while transitioning to a new provider. The airport does not see data collection as critical for their public EV charging but does closely monitor their fleet data for the Airport Carbon Accreditation program. Although SLC is not collecting data on the public chargers currently, SLC staff intend to in the future to help with load management and understanding their remaining charging needs.

## Load Management Considerations and Utility Coordination

Load management has primarily involved working with their partners across the airport to determine overall electrification needs. Rental car companies have started to electrify their fleets

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and SLC is trying to provide capacity to support them where possible. Preparations have included consideration of the capacity on the panels and how many kWs can be provided (as well as how electricity needs for EV charging might affect the overall load on the airport). Load management is less of a concern for publicly available charging but given how it connects to the rest of the airport, SLC must consider the effect of rental company fast chargers on overall operations. The airport has provided capacity for Level 2 charging to the rental car companies but is having to coordinate on the location and capacity to support Level 3 charging equipment.

SLC's annual review of charging needs and potential areas for adding capacity includes an analysis of SLC's power availability across the airport. This analysis involves coordinating with the utility to ensure the available power and infrastructure can support the current and future needs of the airport when adding charging infrastructure. Although SLC has not conducted an electrical capacity assessment yet, there is a plan to do so in the future and this will be coordinated with their utility. The utility requests a 10-year forecast for demand from the airport, which is difficult to estimate given the many aspects of transportation that are electrifying. Emerging electrification, such as electric vertical take-off and landing (eVTOL), would vastly increase SLC's power consumption. Including emerging trends and existing needs, which are growing, into SLC's calculations is difficult. SLC staff noted that they are most comfortable with their 5-year estimates.

## Charging for "Charging" and Other Revenue Considerations

The decision to provide "free" charging for vehicles was driven by city leadership. The city wanted the airport to provide this as an amenity to help incentivize EV ownership and use. Although charging is currently offered for no additional cost, outside of the parking fee, the finance department may revisit that structure in the future. If the cost of providing the amenity continues to grow, SLC may reach a tipping point where they decide to charge for use.

Charging installations have been funded through grants and rebates. The airport has used the Bipartisan Infrastructure Law (BIL) and Zero-Emission Vehicle (ZEV) and Infrastructure Pilot Program from the FAA. The ZEV pilot program allows airports to use Airport Improvement Program (AIP) funds to support ZEV needs. This funding has supported the purchase of EVs for airport use as well as EV chargers. In addition, SLC's local utility offers a rebate program for certain approved charging equipment. The utility will reimburse the airport for 75% of the cost of the station; SLC has used this rebate to support most of their existing charging equipment.

Funding to support the chargers is included in the airport's operations and maintenance budget. The Environmental Programs office has managed to successfully request the necessary funds annually since SLC first started installing chargers. There is no specific line item for EV chargers—the funding comes from the normal operating budget. In terms of the new employee by their facilities team, this was also not specifically designated in the budget for EV charging. This highlights the importance of buy-in from across the airport when installing charging equipment.

## Ensuring Safety

SLC has focused on safety for airside operations but provides some basic safety training about EV charging equipment. This is mostly related to proper cord placement to reduce the chance of an accident or damage to the asset. The airport has not had any incidents in their passenger parking facilities that have caused safety issues.

New installations, especially for DCFC, have included coordination with the fire department. There are fuel dispensers in the facility and so the airport is evaluating the feasibility of providing space for both charging and conventional fuel while ensuring safety. SLC also is coordinating with the fire department on necessary setbacks.

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### Lessons Learned

**1. Understand Service and Management Software Contracts**

SLC staff indicated the importance of understanding the charging provider and the software solution being purchased. Networked chargers and the related service contracts add costs and sometimes this can become too expensive for the charging asset owner to maintain. In addition, understanding how the contract is structured, what is considered proprietary, and what is interoperable is important for success. When increasing their EV charging capacity, airports should not assume that economies of scale will reduce software and data management costs. In addition, "turning off" the charger and not collecting data may disrupt the owner's ability to maintain the asset properly and ensure uptime.

**2. Facility Team Buy-In Assists with Overall Operations**

SLC has benefited from a facilities team (especially their electricians) that has been willing to learn and work with new equipment. The installation and maintenance of their charging assets is kept in-house, which allows SLC greater control and response time to issues. Although leadership initially had concerns about offering free EV charging, they have seen the benefits to the airport and have supported its growth. Buy-in from across the airport has been critical in the continued growth of available charging at SLC.

### Remaining Challenges

Market changes over the past few years have proved challenging and airports, as well as other entities, often struggle with understanding the best charging solution to meet their needs. Ensuring interoperability and transitioning to new software solutions remain a challenge for SLC. However, the support of their internal staff has been crucial to managing their charging effectively.

Overall capacity is another challenge. Power needs for the rental car company operations, a future electrified bus fleet, airside GSE charging, cargo and fire department infrastructure, and increased demand for passenger and employee EV charging creates a large demand. Foreseeing those demands is essential.

## San Francisco International Airport

### Overview

**San Francisco International Airport (SFO)**

City: San Francisco, CA
Hub Size: Large
Enplanements (2022): 20,411,420
EV Chargers: 411 ports (employee and passenger parking)

San Francisco International Airport (SFO) is to the south of the City of San Francisco, California. The airport is owned by the City and County of San Francisco and the parking facilities are privately managed. SFO has four main parking garages and numerous surface lots for their employees with EV charging equipment. SFO first installed EV charging equipment almost a decade ago and currently has 411 charging ports across their passenger and employee parking lots. There are 110 Level 1 chargers (mostly in employee lots but also in long-term parking) and 310 Level 2 charging ports. The Level 2 chargers are in either designated EV parking areas and/or close to the terminal exit or entrance.

### Airport Charging Infrastructure

SFO is installing charging stations to meet demand and to comply with local ordinances and support their sustainability goals. The City and County of San Francisco have an EV readiness

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ordinance that requires 10% of all parking spaces in a facility to provide charging. Given the amount of parking at SFO, this presents an ambitious target. The short-term goal is to install 2,000 chargers with a long-term goal of 6,000. The airport is trying to scale quickly but has encountered workforce challenges that have limited the ability to install chargers rapidly.

In addition to the EV readiness ordinance, SFO has a net-zero carbon emissions goal, which includes reducing their Scope 3 emissions. Scope 3 emissions are sources that SFO influences but does not directly control, such as passenger ground transportation. A 2020 study and roadmap conducted by SFO highlighted that 380,000 metric tons of $CO_2$ at SFO comes from ground transportation and rental cars; this includes passenger, employee, tenant, airline staff, and other airport partner transportation needs. California introduced the Clean Miles Standard as part of state efforts to curb emissions, which requires transportation network companies (TNCs) to electrify by 2030. To meet the needs of TNCs, SFO has released a request for proposals for 14 Direct Current Fast Chargers (DCFCs). There is also the possibility of expanding that site to include 15 fast chargers for use by the public and passengers, but in the short term, the airport is focusing on supporting TNCs. Nevertheless, this charging equipment will be installed within their public parking facilities.

## Selecting and Installing the Charging Equipment

When selecting chargers, the focus has been on Levels 1 and 2 because these meet the use cases of both passenger and employee parking needs at the airport. Demand management is a challenge in the airport environment because passengers will park for extended periods, but charging can be completed in less than a day. SFO has considered different solutions (including valet service for EV charging and mobile charging solutions), but currently has no plan to implement these alternatives. As mentioned, SFO is working toward installing Level 3 chargers/DCFCs to support TNCs and eventually passenger needs. Currently, Level 3 chargers are not required to meet most needs, but with state-level mandates to electrify, SFO is intending to support all users of their parking facilities.

Existing goals and regulations require a rapid increase in available charging, but SFO has run into challenges related to service contracts and longer timelines associated with the city contracting process.

## Managing and Maintaining Charging Equipment

Maintaining their charging assets has been one of the biggest challenges for SFO because of workforce issues. SFO employees do not have the resources to operate, maintain, and repair charging equipment post-installation. Ensuring that existing electricians have the capacity to incorporate EV chargers into their workload is often overlooked, but critical to maintaining operational chargers. SFO is tackling this problem by building capacity into their new parking management contract for charger monitoring and basic maintenance. This change should reduce the demand on their electricians for smaller issues that can be solved by restarting or a quick communication with the charging provider, but more complicated issues will still require SFO electricians to repair them.

Another issue that has compounded the challenges around managing and maintaining charging equipment is the reliability of the data from the chargers. Data from networked chargers is critical to several aspects of management, including load management, but SFO also uses data from chargers to determine maintenance and replacement needs. When information reported from the software conflicts with actual conditions, such discrepancies increase the overall workload and limits decision-making capabilities.

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## Networking and Data Use

Understanding availability, use, and outages is important from a customer service perspective and for load management considerations. Seventy percent of chargers at SFO are networked. Some of the older assets have lower reliability in terms of the data provided, but SFO has mostly installed networked charging equipment. SFO staff have struggled with inconsistent data monitoring capabilities and finding a reliable software and management solution that meets their needs. The airport is working with two providers currently but still struggling with data reliability issues. Although one software solution would be ideal, consistent and reliable data is more important and, so far, SFO has not found one solution that can meet their needs.

Networked chargers and the data provided are critical for load management, especially for larger airports installing increased EV charging capacity. SFO uses the data to assess charger up time and electricity used and how long the charging port is plugged into a vehicle. This information helps SFO staff understand peak demand so that can be included in forecasts and projections communicated to their utility.

## Load Management Considerations and Utility Coordination

SFO staff take an active approach to load management because they are capacity constrained and view this as a critical issue. SFO staff require load management for all EV charger use, including passenger, employee, and fleet charging. Networked charging equipment allows a greater level of control in terms of their electrical draw. For example, California has struggled with grid capacity in recent years and when the airport receives a notice to reduce power consumption significantly, being able to shut off electrical charging capabilities is key. EV charging is not a critical function, so reducing demand there can free power to support core functions.

Handling capacity constraints requires coordination with SFO's utility about the availability of power as well as the possibility for infrastructure upgrades. SFO is looking to add another substation in the future but currently is focused on managing load to work within their constraints The airport forecasts demand campus wide for the next 40 years to inform upgrades to their power systems. The upgrades are a joint venture between the airport and the local utility.

Upgrading electrical infrastructure takes time, so for now, SFO must work within their existing electrical capacity while installing chargers to meet their airport sustainability goals and comply with local ordinances. Networked chargers play a key role in their overall load management, so although non-networked Level 2 chargers are less expensive, the benefits derived from the data and additional control outweigh the cost for SFO.

## Charging for "Charging" and Other Revenue Considerations

SFO currently has Level 1 and Level 2 chargers and does not assess a fee for charging an EV. This policy stems from a triple bottom-line analysis of charging that found that credit sales associated with California's Low Carbon Fuel Standard (LCFS) essentially cover the cost of providing EV charging. If the credit price decreases, SFO may revisit this policy to determine revenue needs. The airport is analyzing the total cost of ownership associated with charging equipment. The results of this analysis will inform the existing and future policies around ownership models and revenue generation.

Although the current policy for Level 1 and Level 2 chargers is not to assess a fee, SFO will be using a "gas station" type model for future Level 3 charging use and the station operator may opt to charge a fee.

SFO has mostly used capital funds as well as some grant funding for their charging equipment. In the future, LCFS credit sales will also be used to support procurement as well as operations

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and maintenance for their assets. Although SFO has been able to access funding for their charging equipment, hiring and training their workforce to support the charging equipment is often an "unseen" cost. Additional workforce capacity is needed at most stages in the process, from contracting to procurement, to the facilities team as well as overall project management. SFO's total cost of ownership study should highlight some of these cost components and assist the airport in the decision-making process. For example, SFO owns, operates, and maintains all its assets at this stage, but this might need to be reconsidered as charging is scaled rapidly at the airport.

## Ensuring Safety

SFO follows building codes and coordinates with their fire department to ensure the safety of EVs within their parking facilities, whether the vehicle is charging or not. Specific building codes relate to EV charging stations in the City and County of San Francisco, including inspection requirements. In addition to following all applicable codes and regulations, the airport separates, where possible, EVs from internal combustion engine vehicles.

In terms of training, SFO does not mandate or have specific training related to EV charging equipment, but licensed electricians are used for all major repair work.

## Lessons Learned

**1. Clear Management Structure**

Clear delineation of responsibilities ensures effective operations and maintenance of charging assets. When initially installing charging equipment, the day-to-day operations and maintenance functions, and where those resided, were not clearly defined. This can reduce efficiency because coordination is required when a charger needs servicing or repairs and without a dedicated employee, responsibility can bounce between different groups. SFO staff recommend understanding the role of different groups in the process, including the Sustainability Office, Facilities Management, and parking management. In addition, determining the reporting order and responsibility for different tasks associated with the charging equipment before installation can smooth the entire ownership process. Ideally, an employee is dedicated to overseeing the charging assets, but this may not be feasible given existing workforce challenges.

**2. Understanding Charging Solutions**

Where possible, understanding the full suite of available charging solutions can ensure that the airport selects the best possible equipment for their needs. Some of SFO's assets are reaching 10 years old and replacements are having to be considered. EV charging solutions have advanced rapidly over the past decade, which can make planning and purchasing decisions difficult to navigate. Ideally, SFO's total cost of ownership analysis for charging equipment will highlight the typical life of the asset along with the cost of repairs and maintenance. As transportation electrification evolves, using data from early adopters should help other airports with their purchase decisions.

## Remaining Challenges

SFO staff mentioned that understanding the tradeoffs in terms of ownership models as well as options for better charging equipment use would be helpful information in building out their EV charging capacity. As SFO develops a strategy to meet goals, what are the benefits of private ownership models? How does airport ownership of assets benefit them from a cost, management, and maintenance perspective? Understanding the experiences of airports with the different ownership models would be useful in planning future installations. Also, the unique parking needs at airports may require innovative charging solutions, but more needs to be understood about the demand and customer willingness to pay for valet service with charging capabilities, mobile charging options, as well as other innovative solutions.

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*Abbreviations and acronyms used without definitions in TRB publications:*

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| A4A | Airlines for America |
| AAAE | American Association of Airport Executives |
| AASHO | American Association of State Highway Officials |
| AASHTO | American Association of State Highway and Transportation Officials |
| ACI–NA | Airports Council International–North America |
| ACRP | Airport Cooperative Research Program |
| ADA | Americans with Disabilities Act |
| APTA | American Public Transportation Association |
| ASCE | American Society of Civil Engineers |
| ASME | American Society of Mechanical Engineers |
| ASTM | American Society for Testing and Materials |
| ATA | American Trucking Associations |
| CTAA | Community Transportation Association of America |
| CTBSSP | Commercial Truck and Bus Safety Synthesis Program |
| DHS | Department of Homeland Security |
| DOE | Department of Energy |
| EPA | Environmental Protection Agency |
| FAA | Federal Aviation Administration |
| FAST | Fixing America's Surface Transportation Act (2015) |
| FHWA | Federal Highway Administration |
| FMCSA | Federal Motor Carrier Safety Administration |
| FRA | Federal Railroad Administration |
| FTA | Federal Transit Administration |
| GHSA | Governors Highway Safety Association |
| HMCRP | Hazardous Materials Cooperative Research Program |
| IEEE | Institute of Electrical and Electronics Engineers |
| ISTEA | Intermodal Surface Transportation Efficiency Act of 1991 |
| ITE | Institute of Transportation Engineers |
| MAP-21 | Moving Ahead for Progress in the 21st Century Act (2012) |
| NASA | National Aeronautics and Space Administration |
| NASAO | National Association of State Aviation Officials |
| NCFRP | National Cooperative Freight Research Program |
| NCHRP | National Cooperative Highway Research Program |
| NHTSA | National Highway Traffic Safety Administration |
| NTSB | National Transportation Safety Board |
| PHMSA | Pipeline and Hazardous Materials Safety Administration |
| RITA | Research and Innovative Technology Administration |
| SAE | Society of Automotive Engineers |
| SAFETEA-LU | Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (2005) |
| TCRP | Transit Cooperative Research Program |
| TEA-21 | Transportation Equity Act for the 21st Century (1998) |
| TRB | Transportation Research Board |
| TSA | Transportation Security Administration |
| U.S. DOT | United States Department of Transportation |

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